growth, (ii) activation of innate immune mechanisms (including inflammation), and (iii) control of bone marrow hematopoiesis (cf. figure 10.4).

• Cytokines may act sequentially, through one cytokine inducing production of another or by transmodulation of the receptor for another cytokine; they can also act synergistically or antagonistically.

• Their roles *in vivo* can be assessed by gene 'knockout', transfection or inhibition by specific antibodies.

Different T-cell subsets can make different cytokines

• As immunization proceeds, Th tend to develop into two subsets: Th1 cells concerned in inflammatory processes, macrophage activation and delayed sensitivity make IL-2 and -3, IFNγ, TNF, lymphotoxin and GM-CSF; Th2 cells help B-cells to synthesize antibody and secrete IL-3, -4, -5, -6 and -13, TNF and GM-CSF. IL-10 is secreted by Th2 cells in mice but by both Th1 and Th2 subsets in humans.

• Early interaction of antigen with macrophages or dendritic cells producing IL-12 or with a variety of cells secreting IL-4 will skew the responses to Th1 or Th2, respectively.

• Other subsets may exist, including TGFβ-secreting Th3 (Tr1) regulatory cells.

• Tc1 (IFNγ) and Tc2 (IL-4) populations can also be distinguished.

Activated T-cells proliferate in response to cytokines

• IL-2 acts as an autocrine growth factor for Th1 and paracrine for Th2 cells which have upregulated their IL-2 receptors.

• Cytokines act on cells which express the appropriate cytokine receptor.

T-cell effectors in cell-mediated immunity

• Cytokines mediate chronic inflammatory responses and induce the expression of MHC class II on endothelial cells, a variety of epithelial cells and many tumor cell lines, so facilitating interactions between T-cells and nonlymphoid cells.

• Differential expression of chemokine receptors permits selective recruitment of neutrophils, macrophages, dendritic cells and T- and B-cells.

• TNF synergizes with IFNγ in killing cells.

• Cytotoxic T-cells are generated against cells (e.g. virally infected) which have intracellularly derived peptide associated with surface MHC class I. They kill using lytic granules containing perforin, granzymes and TNF.

• T-cell-mediated inflammation is strongly downregulated by IL-4 and IL-10.

Proliferation of B-cell responses is mediated by cytokines

• Early proliferation is mediated by IL-4 which also aids IgE synthesis.

• IgA producers are driven by TGFβ and IL-5.

• IL-4 plus IL-5 promote IgM and IL-4, -5, -6 and -13 plus IFNγ stimulate IgG synthesis.

Events in the germinal center

• There is clonal expansion, isotype switch and mutation in the dark zone centroblasts.

• The B-cell centroblasts die through apoptosis unless rescued by certain signals which upregulate *bcl-2*. These include cross-linking of surface Ig by complexes on follicular dendritic cells and engagement of the CD40 receptor which drives the cell to the memory compartment.

• The selection of mutants by antigen guides the development of high affinity B-cells.

The synthesis of antibody

• RNA for variable and constant regions is spliced together before leaving the nucleus.

• Differential splicing allows coexpression of IgM and IgD with identical V regions on a single cell and the switch from membrane-bound to secreted IgM.

Ig class switching occurs in individual B-cells

• IgM produced early in the response switches to IgG, particularly with thymus-dependent antigens. The switch is largely under T-cell control.

• IgG, but not IgM, responses improve on secondary challenge.

Antibody affinity during the immune response

• Low doses of antigen tend to select high affinity B-cells and hence antibodies since only these can be rescued in the germinal center.

• For the same reasons, affinity matures as antigen concentration falls during an immune response.

Memory cells

• Memory T-cells can be maintained in the absence of antigen.

• However, immune complexes on the surface of follicular dendritic cells in the germinal centers provide a long-term source of antigen.

• Memory cells have higher affinity than naive cells, in the case of B-cells through somatic mutation, and in the case of T-cells through selective proliferation of cells with higher affinity receptors and through upregulated expression of associated molecules such as CD2 and LFA-1, which in-

(*continued*)

crease the avidity (functional affinity) for the antigen-presenting cell.

- Activated memory and naive T-cells are distinguished by the expression of CD45 isoforms, the former having the CD45RO phenotype, the latter CD45RA. It seems likely that a proportion of the CD45RO population reverts to a CD45RA pool of resting memory cells. $CD45RA^-$ memory cells can be divided into $CCR7^+$ central memory and $CCR7^-$ effector memory cells.
- High levels of CD44 expression are also characteristic of memory T-cells, low level expression being associated with naive T-cells.

See the accompanying website (**www.roitt.com**) for multiple choice questions.

FURTHER READING

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CHAPTER 11

Control mechanisms

INTRODUCTION

The acquired immune response evolved so that it would come into play when contact with an infectious agent is made. The appropriate antigen-specific cells expand, often to form a sizable proportion of the lymphocytes in the local lymphoid tissues, the effectors eliminate the antigen and then the response quietens down and leaves room for reaction to other infections. Feedback mechanisms must operate to limit the response; otherwise, after antigenic stimulation, we would become overwhelmed by the responding clones of T-cells and antibody-forming cells and their products—obviously an unwelcome state of affairs, as may be clearly seen in multiple myeloma, where control over lymphocyte proliferation is lost. It makes sense for **antigen to be a major regulatory factor** and for lymphocyte responses to be driven by the presence of antigen, falling off in intensity as the antigen concentration drops (figure 11.1). There is abundant evidence to support this view. Antigens can stimulate the proliferation of specific lymphocytes *in vitro*. Clearance of antigen *in vivo* by injection of excess antibody during the course of an immune response leads to a dramatic drop in antibody synthesis and the number of antibody-secreting cells.

ANTIGENS CAN INTERFERE WITH EACH OTHER

The presence of one antigen in a mixture of antigens can drastically diminish the immune response to the others. This is true even for epitopes within a given molecule; for example, the response to epitopes on the Fab fragment of IgG is far greater when the Fab rather than whole IgG is used for immunization due to the inhibitory nature of the Fc region. Factors which determine immunodominance include the precursor frequency of the B-cells bearing antigen receptors for different epitopes on the antigen, the relative affinity of these antigen receptors for their respective epitopes, the degree to which the surface membrane antibody protects the epitope from proteolysis following internalization of the antibody–antigen complex, and the level of competition of processed antigenic peptides for the *m*ajor *h*istocompatibility *c*omplex (MHC) groove. There is a clear hierarchy of epitopes with respect to this competitive binding based on differential accessibility to proteases as the molecule unfolds, and the presence or absence of particular amino acid sequences which facilitate breakdown to yield peptides in high abundance and with relatively high affinity for the MHC (figure 11.2). Thus, Sercarz envisages **dominant epitopes**, which bag the lion's share of the available MHC grooves, **subdominant epitopes**, which are less successful, and **cryptic epitopes**, which generate

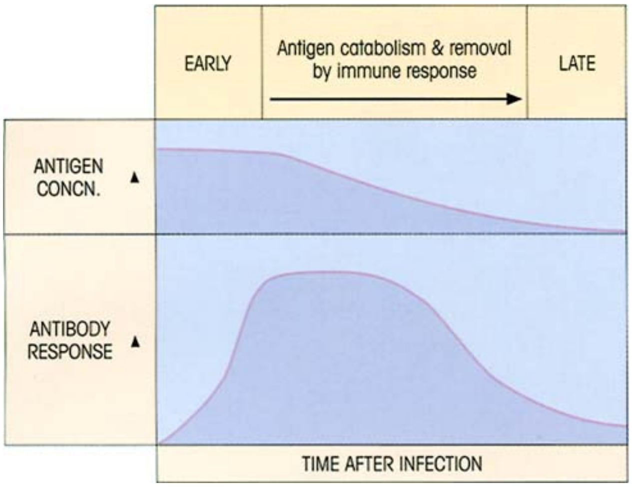

Figure 11.1. Antigen drives the immune response. As antigen concentration falls due to catabolism and elimination by antibody, the intensity of the immune response declines, but is maintained for some time at a lower level by antigen trapped on germinal center follicular dendritic cells.

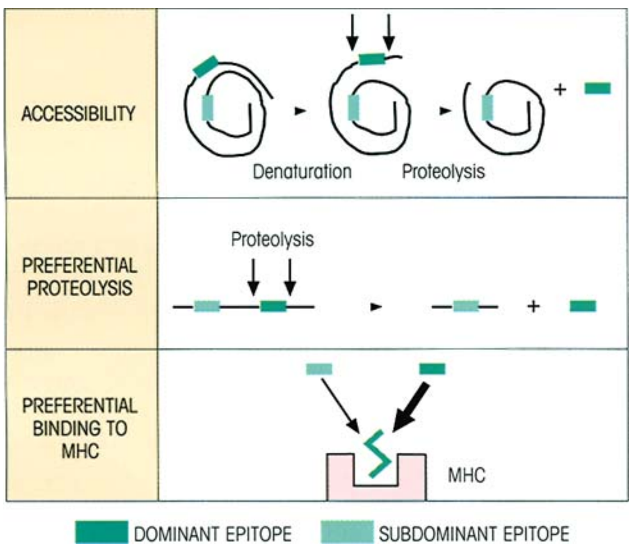

Figure 11.2. Mechanisms of epitope dominance at the MHC level. The other factor which can influence dominance is the availability of reactive T-cells; if these have been eliminated, e.g. through tolerization by cross-reacting self-antigens, a peptide which may have dominated the MHC groove would be unable to provoke an immune response.

miserably low concentrations of peptide–MHC that are ignored by potentially reactive naive T-cells.

Clearly, the possibility that certain antigens in a mixture, or particular epitopes in a given antigen, may block a desired protective immune response has obvious implications for vaccine design. Contrariwise, the identification of inhibitory peptides with a predatory affinity for the MHC groove(s) should provide therapeutic agents to quash unwanted hypersensitivity reactions.

COMPLEMENT AND ANTIBODY ALSO PLAY A ROLE

Innate immune mechanisms are usually first on the scene and activation of the alternative pathway of complement activation will lead to C3d deposition on the microbe. When C3d-coated antigens are recognized by the B-cell, cross-linking of the BCR and the CD21 complement receptor, with its associated signal-transducing molecule CD19, enhances B-cell activation (figure 11.3a). In contrast, cross-linking of the BCR with FcγRIIB1 (cf. p. 50) delivers a negative signal by suppressing tyrosine phosphorylation of CD19 (figure 11.3b). Thus, removal of circulating antibody by plasmapheresis during an ongoing response leads to an increase in synthesis, whereas injection of preformed IgG antibody markedly hastens the fall in the number of antibody-forming cells (figure 11.4) consistent with feedback control on overall synthesis.

In complete contrast, injection of IgM antibodies enhances the response (figure 11.4), presumably by cross-linking antigen bound to the sIgM receptors without activating the Fcγ inhibitory receptor, and perhaps also via the generation of C3d by the classical pathway of complement activation. Since antibodies of this isotype are either already present amongst the broadly reactive natural antibodies, or if not will certainly appear at an early stage after antigen challenge, they would be useful in boosting the initial response.

ACTIVATION-INDUCED CELL DEATH

Although clearance of antigen from the body by the immune system will clearly lead to a downregulation of lymphocyte proliferation due to the absence of a signal through the antigen receptor, even in the presence of antigen the signals provided do not lead to the continuous proliferation of cells, but rather set off a train of events that, unless the cells are protected in some way, leads to *a*ctivation-*i*nduced *c*ell *d*eath (AICD) by apoptosis. It seems that AICD is perhaps primarily a feature of CD4 cells, allowing these cells to be cleared from the system following their proliferation. Subsequent to activation, T-cells upregulate death receptors and their ligands. If the ligands remain associated with the cell surface they can activate apoptosis in adjacent cells. However, they are often released from the cell surface by proteases, producing soluble forms which in some cases retain activity, for example the soluble version of the TNF-related apoptosis-inducing ligand (TRAIL)

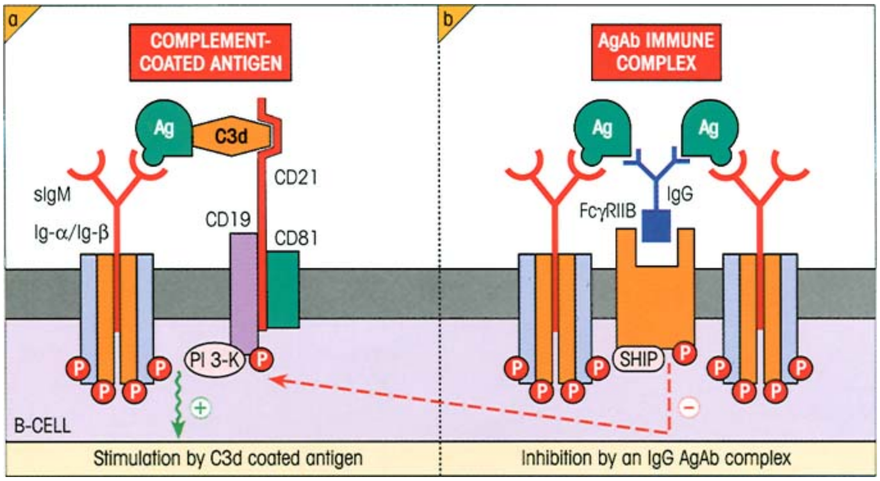

Figure 11.3. Cross-linking of surface IgM antigen receptor to the CD21 complement receptor stimulates, and to the Fcγ receptor FcγRIIB1 inhibits, B-cells. (a) Following activation of complement, C3d becomes covalently bound to the microbial surface. The CD21 complement receptor binds C3d and signals through its associated CD19 molecule. The CD19, CD21 and CD81 (TAPA-1) molecules form the B-cell coreceptor and cross-linking of this complex to the surface IgM of the BCR leads to tyrosine phosphorylation of CD19 and subsequent binding of phosphatidylinositol 3-kinase (PI3-K), leading to B-cell activation. (b) The FcγRIIB1 molecule possesses a cytoplasmic immunoreceptor tyrosine-based inhibitory motif (ITIM) and, upon cross-linking to membrane Ig, becomes phosphorylated and binds the inositol polyphosphate 5′-phosphatase SHIP. This suppresses phosphorylation of CD19 and thus inhibits B-cell activation.

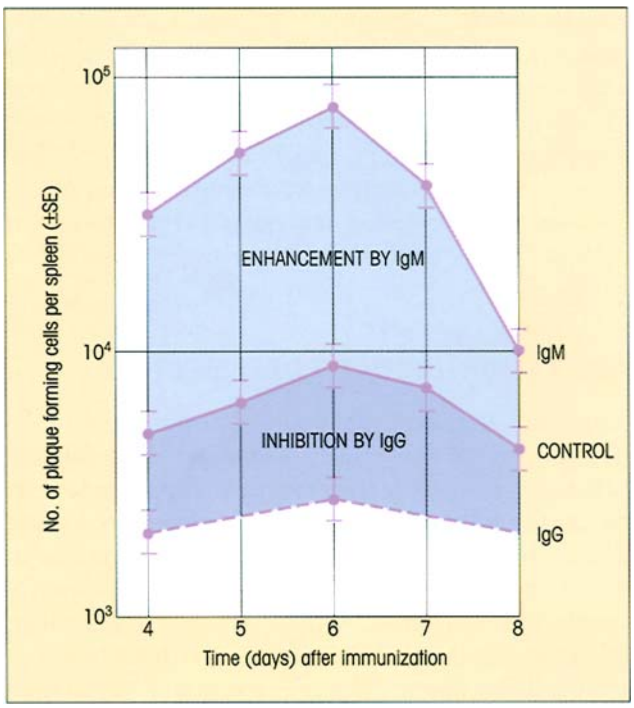

Figure 11.4. Time-course of enhancement of antibody response to sheep red blood cells (SRBC) due to injection of preformed IgM, and of suppression by preformed IgG antibodies. Mice received monoclonal IgM anti-SRBC, IgG anti-SRBC or medium alone intravenously 2 hours prior to immunization with 10^5 SRBC. (Data provided by J. Reiter, P. Hutchings, P. Lydyard and A. Cooke.)

retains the ability to signal through the receptor TRAIL-R1. Such soluble ligands can potentially mediate either paracrine or autocrine cell death. The death receptors are members of the tumor necrosis factor receptor (TNF-R) family and include TNFRI, CD95 (Fas), TRAMP (TNF receptor apoptosis-mediating protein), the aforementioned TRAIL-R1, TRAIL-R2 and death receptor (DR) 6. Apoptosis induction through these receptors initially involves cleavage of the inactive cysteine protease procaspase 8 to yield active caspase 8. Ultimately, this activation pathway converges with the apoptosis pathway induced by cellular stress, both leading to the activation of downstream effector caspases (figure 11.5). In addition to the death receptors, there are also a number of 'decoy receptors' which bind the potentially apoptosis-inducing ligands but do not signal.

When initially activated by peptide–MHC, T-cells are resistant to apoptosis but become progressively sensitive. A number of molecules are known to be protective against apoptosis; for example, bcl-2 and bcl-X_L, which appear to act as watchdogs preventing the release of pro-apoptotic proteins from the mitochondria. Of particular relevance to death receptor-mediated AICD, however, is the molecule FLIP (FLICE inhibitory protein, FLICE being an older name for caspase 8). FLIP bears structural similarity to caspase 8, and therefore by competitive inhibition prevents recruitment of this caspase into the *d*eath-*i*nducing *s*ignaling *c*omplex (DISC) (figure 11.5). It appears that the

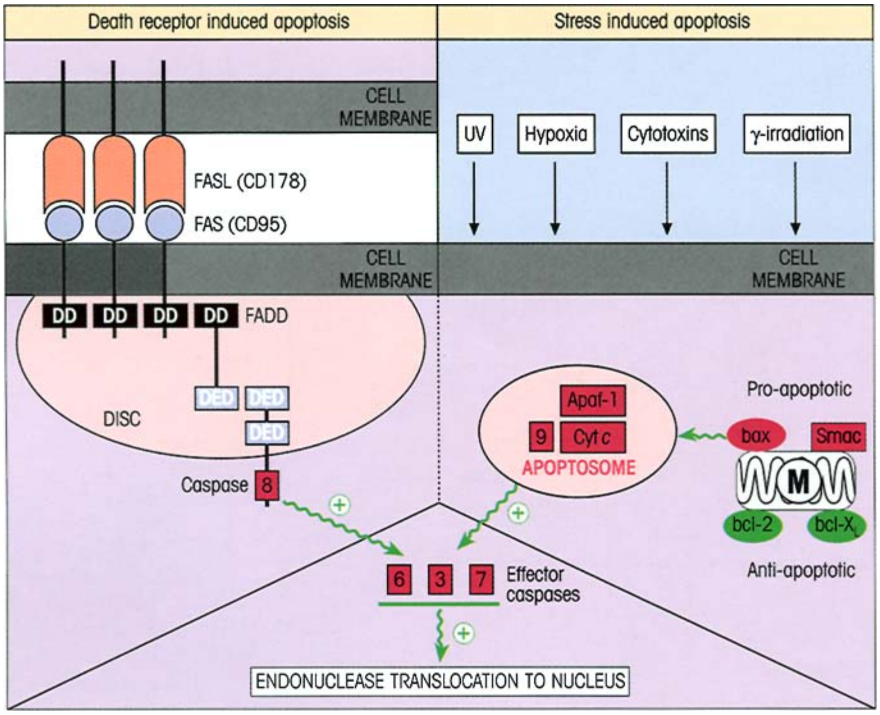

Figure 11.5. Activation-induced cell death. Receptor-based induction of apoptosis involves the trimerization of TNF-R family members (for example, Fas) by trimerized ligands (for example, Fas-ligand). This brings together cytoplasmic death domains (DD) which can recruit a number of death effector domain (DED)-containing adaptor molecules to form the death-inducing signaling complex (DISC). The different receptors use different combinations of DED-containing adaptors; Fas uses FADD (*F*as-*a*ssociated protein with *d*eath *d*omain). The DISC induces the cleavage of inactive procaspase 8 into active caspase 8, with subsequent activation of downstream effector caspases. This process eventually leads to the release of the endonuclease known as caspase-activated DNase (CAD) from a restraining protein (inhibitor of CAD; ICAD) in the cytoplasm, with subsequent translocation of the endonuclease to the nucleus. A second pathway of apoptosis induction, often triggered by cellular stress, involves a number of mitochondria-associated proteins including cytochrome *c*, Smac/DIABLO and the bcl-2 family member bax. Caspase 9 activation is the key event in this pathway and requires association of the caspase with a number of other proteins including the cofactor Apaf-1; the complex formed incorporates cytochrome *c* and is referred to as the apoptosome. The activated caspase 9 then cleaves procaspase 3. Although the death receptor and mitochondrial pathways are shown as initially separate in the figure, there is some cross-talk between them. Thus, caspase 8 can cleave the bcl-2 family member bid (not shown), a process which promotes cytochrome *c* release from mitochondria. Other members of the bcl-2 family, such as bcl-2 itself and bcl-X_L, inhibit apoptosis, perhaps by preventing the release of pro-apoptotic molecules from the mitochondria. M, mitochondrion.

balance between caspase 8 and FLIP levels can determine the fate of the cell when the death receptor is engaged by its ligand, but does not affect apoptosis induced by the stress-activated mitochondrial pathway (figure 11.6).

T-CELL REGULATION

T-helper cells

There is abundant evidence to suggest that different populations of Th cells are specialized for different helper functions. With respect to help for antibody production, T-cell lines derived from Peyer's patches are much better at helping IgA-producing B-cells from Peyer's patch precursors than are splenic T-cell lines. The help is for IgA-precommitted B-cells rather than for induction of the class switch to IgA, since Peyer's patch T-cells do not markedly enhance IgA production by splenic B-cells. Evidence for the production of an IgE-binding factor from Fcε receptor-bearing T-cells, which enhances B-cell secretion of IgE, has been obtained; intriguingly, a 13 kDa T-cell-derived cytokine called *g*lycosylation *i*nhibiting *f*actor (GIF) inhibits *N*-glycosylation of the IgE-binding factor so that it now becomes a suppressor of IgE synthesis. It should also be evident from the previous discussion of AICD that Th cells will not be around to expand B-cell and Tc clone sizes indefinitely.

T-cell suppression

It is perhaps inevitable that nature, having evolved a functional set of T-cells which promote immune

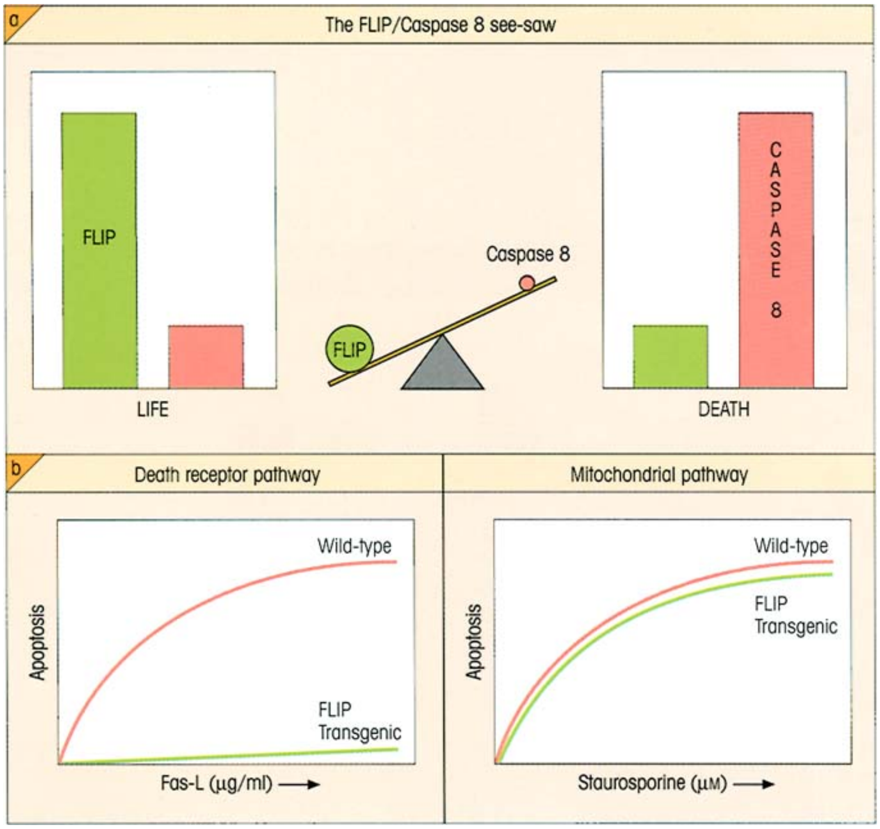

Figure 11.6. Life and death decisions. (a) The relative amounts of anti-apoptotic FLIP and pro-apoptotic caspase 8 can determine the fate of the cell. (b) Experiments involving overexpression of FLIP in a transgenic mouse model show that this protein protects T-cells from AICD stimulated through the death receptor pathway by Fas-ligand, but not from cell death triggered via the mitochondrial pathway using the drug staurosporine. (Based on data obtained by J. Tschopp and colleagues.)

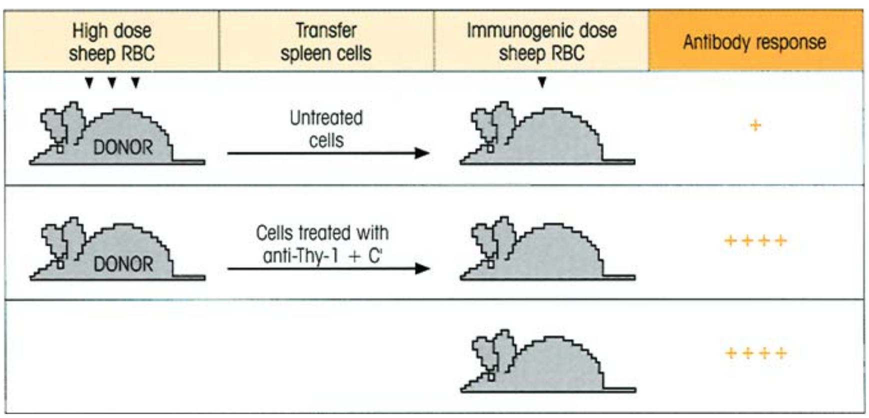

Figure 11.7. Demonstration of T-suppressor cells. A mouse of an appropriate strain immunized with an immunogenic dose of sheep erythrocytes makes a strong antibody response. However, if spleen cells from a donor of the same strain previously injected with a high dose of antigen are first transferred to the syngeneic animal, they depress the antibody response to a normally immunogenic dose of the antigen. The effect is lost if the spleen cells are first treated with a T-cell-specific antiserum (anti-Thy-1) plus complement, showing that the suppressors are T-cells. (After Gershon R.K. & Kondo K. (1971) *Immunology* **21**, 903.)

responses, should also develop a regulatory set whose job would be to modulate the helpers. T-cell suppression was first brought to the serious attention of the immunological fraternity by a phenomenon colorfully named by its discoverer, 'infectious tolerance'. Quite surprisingly it was shown that, if mice were made unresponsive by injection of a high dose of *s*heep *r*ed *b*lood *c*ells (SRBC), their T-cells would suppress specific antibody formation in normal recipients to which they had been transferred (figure 11.7). It may not be apparent to the reader why this result was at all surprising, but at that time antigen-induced tolerance was regarded essentially as a negative phenomenon involving the depletion or silencing of clones rather than a state of active suppression. Over the years, T-cell suppression has been shown to modulate a variety of humoral and cellular responses, the latter including delayed-type hypersensitivity, cytotoxic T-cells and antigen-specific T-cell proliferation. However, the existence of dedicated professional T-suppressor cells is a question which has generated a great deal of heat.

Suppressor and helper epitopes can be discrete

Detailed analysis of murine responses to antigens such as hen egg-white lysozyme tells us that certain determinants can evoke very strong suppressor rather than helper responses depending on the mouse strain, and also that T-suppressors directed to one determinant can switch off helper and antibody responses to other determinants on the same molecule. Thus mice of *H-2^b* haplotype respond poorly to lysozyme because they develop dominant suppression; however, if the three N-terminal amino acids are removed from the antigen, these mice now make a splendid response, showing that the T-suppression directed against the determinant associated with the N-terminal region has switched off the response to the remaining determinants on the antigen. Similar results have been obtained in several other systems. This must imply that the antigen itself acts as a form of bridge to allow communication between T-suppressor and cells reacting to the other determinants, as might occur through these cells binding to an antigen-presenting cell expressing several different processed determinants of the same antigen on its surface (figure 11.8).

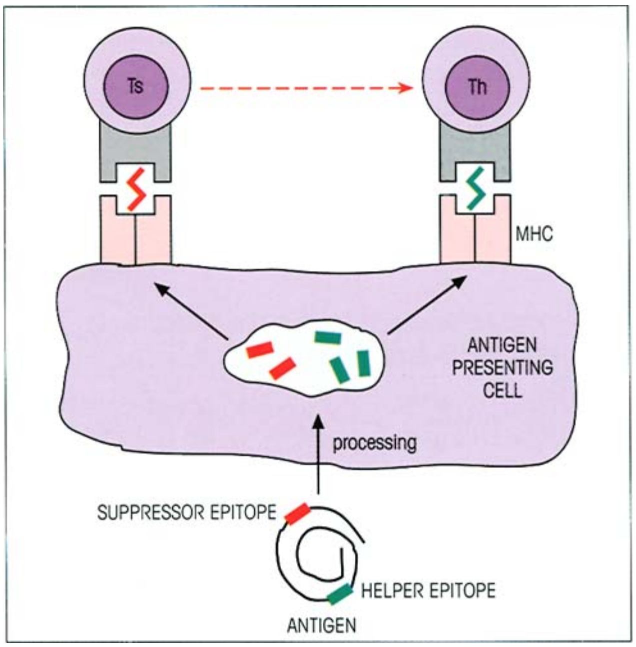

Figure 11.8. Possible mechanism to explain the need for a physical linkage between suppressor and helper epitopes. The helper and putative suppressor cells can interact by binding close together on the surface of an antigen-presenting cell, which processes the antigen and displays the different epitopes on separate MHC molecules on its surface.

Characteristics of suppression

What of the effectors of suppression? In a number of experimental systems the cells which mediate suppression are far more vulnerable than helpers to adult thymectomy, X-irradiation and cyclophosphamide. For example, adult thymectomy in the mouse has little effect on the Th population, but leads to a fall in the T-suppressors, thereby increasing the response to T-independent antigens and preventing the fall-off in IgE antibody to haptens coupled with *Ascaris* extracts which occurs in intact animals.

Classically, suppressor T-cells have been described as CD8$^+$. For example, the B10.A (2R) mouse strain has a low immune response to lactate dehydrogenase β (LDHβ) associated with the possession of the *H-2Eβ* gene of *k* rather than *b* haplotype. Lymphoid cells taken from these animals after immunization with LDHβ proliferate poorly *in vitro* in the presence of antigen, but if CD8$^+$ cells are depleted, the remaining CD4$^+$ cells give a much higher response. Adding back the CD8$^+$ cells reimposes the active suppression. Human suppressor T-cells can also belong to the CD8 subset. Thus, CD8$^+$ CD28$^-$ cells can prevent antigen-presenting B-cells from upregulating costimulatory B7 molecules in response to CD40-mediated signals from the CD40 ligand on Th cells. Following interaction with the CD8 T-cells, the APCs are then capable of inducing anergy in Th cells. This effect on B7 is mediated by inhibition of NFκB activation in the APC, an event necessary for transcription of both the B7.1 (CD80) and B7.2 (CD86) molecules.

More recently, it has become appreciated that regulatory CD4$^+$ cells can also be major effectors of suppression. In a model of *i*nflammatory *b*owel *d*isease (IBD), in which transfer of CD45RB^{high}CD4$^+$ T-cells from normal donor mice into mice with *s*evere *c*ombined *i*mmuno*d*eficiency (SCID) leads to the development of IBD, the disease can be prevented if CD45RB^{low}CD4$^+$ regulatory T-cells (Tr) are transferred at the same time. This immunoregulatory effect could be blocked by antibodies to the IL-10 receptor or to TGFβ in mice, or to IL-4 and TGFβ in rats.

Suppressor cells of the CD4 phenotype have also been identified within the minor population which express the IL-2 receptor α chain (CD25). These suppressors are able to abrogate the responses of CD4$^+$ CD25$^-$ T-cells by an as yet undefined cell contact-dependent, cytokine-independent mechanism.

Suppression is a regulated phenomenon

Let us look in a little more depth at some potential

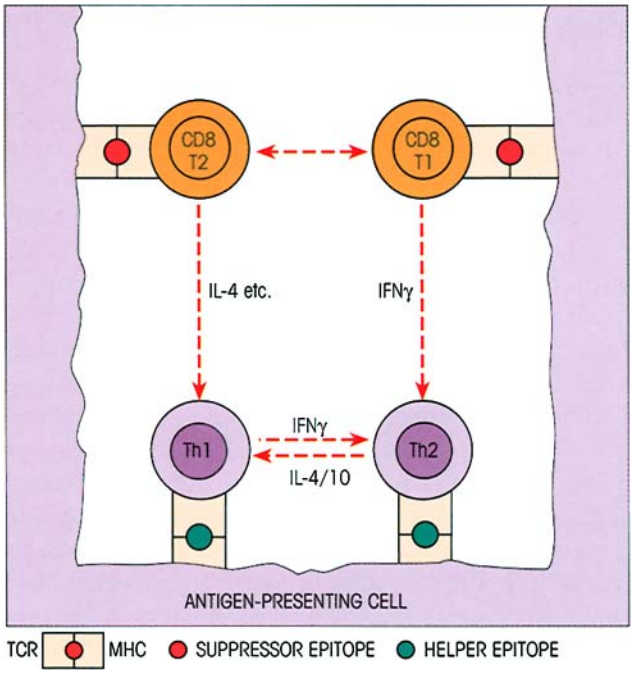

Figure 11.9. Mutual antagonisms between T-cell subsets linked indirectly by processed antigen on an antigen-presenting cell lead to functionally distinct modes of suppression. (Leaning heavily on Bloom B.R., Salgame P. & Diamond B. (1992) *Immunology Today* **13**, 131.) Yet another mechanism may prove to be important. Unlike the mouse, many other mammalian species can express MHC class II on a proportion of their activated T-cells; presentation of processed peptide by these cells can induce CD4-positive cytotoxic cells with suppressor potential. We also need to know more about the circumstances leading to the production of TGFβ by suppressors since this cytokine inhibits T-cell proliferation.

mechanisms of suppression. We have already entertained the idea that antigen-linked T–T interactions can occur on the surface of an antigen-presenting cell (figure 11.8) and the concept of T1 and T2 CD8 subsets paralleling the Th1/Th2 dichotomy. Furthermore, the evidence for mutual antagonism (suppression) between Th1 and Th2 cells is very strong indeed. One could postulate downregulation of Th1 cells by type 2 IL-4-producing CD8 cells, and suppression of Th2 cells by type 1 IFNγ-producing CD8 cells, interacting on the surface of an antigen-presenting cell (figure 11.9). In this model, when the immune response has locked onto a particular mode, e.g. Th1-mediated cellular immunity, other types of response, such as T–B collaboration, are restricted through a cytokine inhibitory effect. Although these cells mediate T-suppression, they would not be called dedicated professional suppressors since, in a sense, their suppressive powers are a by-product of their main defensive function. Perhaps we need these cytokine-secreting Tc cells to prevent Th cells getting out of hand by excessive proliferation, just as IgG holds back the B-cells by feedback control. There is indeed abundant evidence for such a regulatory mechanism whereby suppressors can be induced by Th1 cells.

The nature of the antigen-presenting cell, upon whose surface these interactions are thought to occur, may influence the type of cells involved and the outcome. It may be recalled that dendritic cells, macrophages, B-cells, activated T-cells and even epithelial cells with upregulated class II can all act as antigen presenters. Of interest is the finding that irradiation with UVB light inactivates dendritic cells and stimulates T-suppression; IL-2 and IFNγ are downregulated and IL-4 production is increased.

Other issues which require solutions are the mechanism by which high-dose antigen induces suppression, the role of cytotoxic CD4 and CD8 T-cells in these phenomena and the extent to which idiotype-specific T-suppressors contribute.

It is important not to lose track of the so-called 'natural suppressor cells', such as those provoked by total lymph node irradiation, and we would like to draw attention to a report that natural killer (NK) cells can inhibit the one-way mixed lymphocyte reaction (see p. 352) or the primary IgM response to sheep cells *in vitro*, by suppressing dendritic cells which have already taken up the antigen. This offers a further opportunity for feedback control since IFNγ and IL-2 produced by Th can activate NK cells.

Overall, suppression may be most commonly mediated by secretion of cytokines such as TGFβ, by secretion of yet to be defined suppressor molecules and by inactivation of APC function. Make no mistake, suppression is still a murky area, and is not for the unwary, but the light at the end of the tunnel is beckoning.

Effector T-cells are guided to the appropriate target by MHC surface molecules

Not only do MHC molecules bearing a tell-tale peptide signal the presence of an intracellular precursor (cf. p. 95, Chapter 5), they also ensure that the T-cell makes contact with the surface of the appropriate target cell.

Let us explore this point by looking at the role of Tc in viral infection. When a cell is first infected with virus, there is an eclipse phase during which the machinery of the cell is being switched for viral replication and the only marker of the complete microbe is the processed viral antigen peptide on the cell surface. At this stage, killing of the cell by a cytotoxic T-cell will prevent viral replication. The killer

T-cell knows it has reached its target when it recognizes the surface viral peptide in association with class I MHC molecules which are present on nearly every cell in the body. Thus the killer cell operates on the basis that processed viral antigen is the code for '**viral infection**' and class I molecules are the code for '**cell**' (cf. figure 5.26).

The situation is quite different with intracellular bacteria and protozoa which do not go through an eclipse phase after phagocytosis by macrophages, but are held as infectious entities; lysis by cytotoxic T-cells will merely release the organisms, not kill them. A separate strategy is required and, in this case, the effector Th1 lymphocyte recognizes the infected macrophage by the presence of microbial antigen on the surface in association with a class II molecule which is now a code for 'macrophage'. This interaction triggers the release of macrophage-stimulating cytokines, dominant among which is IFNγ. Multiple microbicidal mechanisms are triggered in the activated macrophages, including the formation of reactive oxygen intermediates and the synthesis of NO·, to the detriment of the intracellular parasites (see p. 265). Similarly, in T–B cooperation, the **B-cell** is recognized through its class II molecule associated with the foreign antigen, although in this case costimulatory signals through CD40L–CD40 interactions are required for activation. T-cells mediating **suppression** also utilize MHC molecules, but the mechanisms are unclear. In summary, each antigen-specific T-lymphocyte subset has to communicate with a particular cell type in order to make the appropriate immune response, and it does so by recognizing not only processed foreign antigen but also the particular MHC molecule used as a marker of that cell (table 11.1).

Some types of immunoregulatory cells, such as certain γδ T-cells, may recognize activation-induced, nonclassical MHC molecules, as appears to be the case with the recognition of the class Ib molecules T22 and T10 in the mouse, although the precise role of such cells is still being elucidated.

Table 11.1. Guidance of T-subpopulations to appropriate target cell by MHC molecules.

Function	Cell interaction	MHC marker on target cell
T-cell priming	T–dendritic cell	II (+*B7/CD28)
T-helper (Th1)	T–macrophage	II (+peptide)
T-helper (Th2)	T–B	II (+*CD40L/CD40)
T-suppressor	T–T	?
T-cytotoxic	T–target cell	I

*Costimulatory molecules and their ligand.

IDIOTYPE NETWORKS

Jerne's network hypothesis

The hypervariable loops on the immunoglobulin molecule which form the antigen combining site have individual characteristic shapes which can be recognized by the appropriate antibodies as idiotypic determinants (cf. p. 42). There are hundreds of thousands, if not more, different idiotypes in one individual.

Jerne reasoned brilliantly that the great diversity of idiotypes would to a considerable extent mirror the diversity of antigenic shapes in the external world. Thus, he said, if lymphocytes can recognize a whole range of foreign antigenic determinants, they should be able to recognize the idiotypes on the antigen receptors of other lymphocytes. They would therefore form a large network or series of networks depending upon **idiotype–anti-idiotype recognition** between T- and B-cells (figure 11.10), and the response to an external antigen perturbing this network would be conditioned by the state of the idiotypic interactions.

Evidence for idiotypic networks

Anti-idiotype can be induced by autologous idiotypes

There is no doubt that the elements which can form an idiotypic network are present in the body, and autoanti-idiotypes occur during the course of antigen-induced responses. For example, certain strains of mice injected with pneumococcal vaccines make an antibody response to the phosphorylcholine groups in which the germ-line-encoded idiotype T15 dominates. If the individual antibody-forming cells are examined by plaque assays at different times after immunization, waves of $T15^+$ and of anti-T15 (i.e. autoanti-idiotype) cells are demonstrable. Similarly, immunization with the acetylcholine agonist BISQ, followed by fusion of the spleen cells to produce hybridomas, yielded a series of anti-BISQ monoclonals (idiotypes) and a smaller number of anti-idiotypic monoclonals, of which a surprising proportion behaved as internal images of BISQ in their ability to stimulate acetylcholine receptors (figure 11.11). Anti-idiotypic reactivity has also been demonstrated in T-cell populations using various experimental systems. Indeed, anti-idiotypic T-cells recognizing peptide derived from the CDR2, CDR3 and framework regions of other TCRs appear to form a part of the normal human T-cell repertoire. Immunization of multiple sclerosis patients with myelin basic protein reactive T-cell clones induces $CD8^+$ cytotoxic T-cells which recognize a peptide

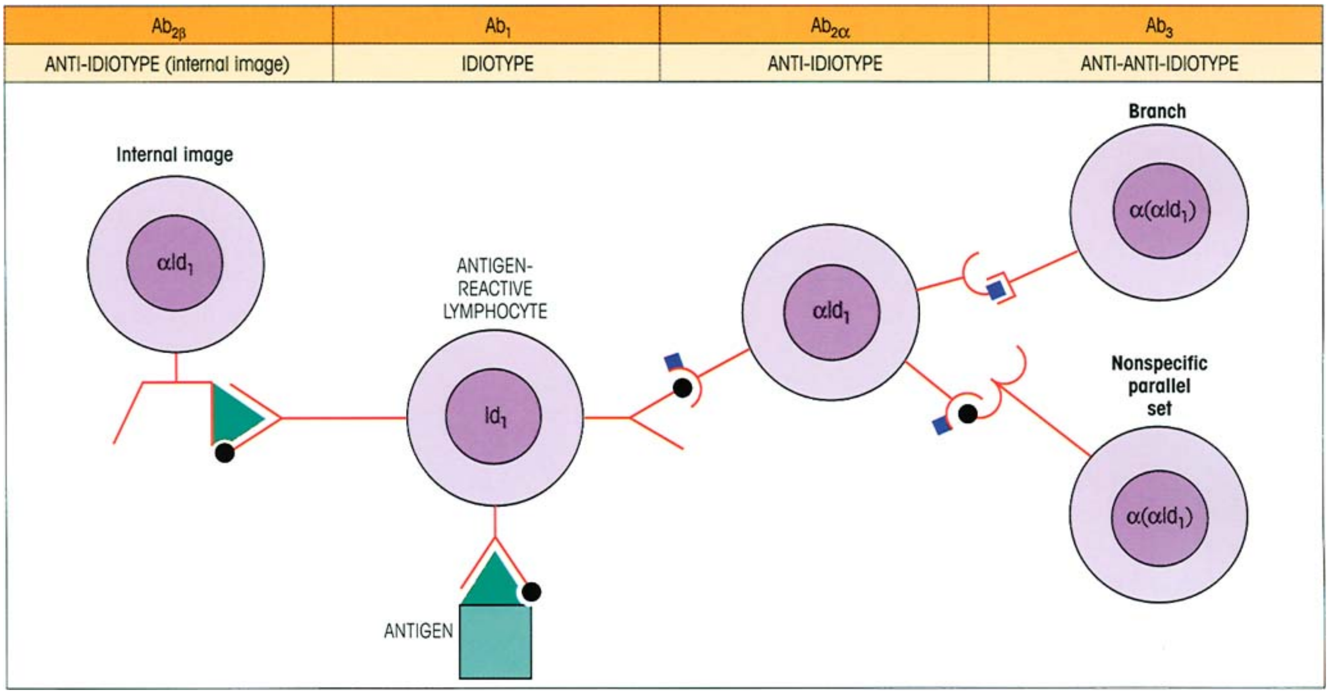

Figure 11.10. Elements in an idiotypic network in which the antigen receptors on one lymphocyte reciprocally recognize an idiotype on the receptors of another. T-helper, T-suppressor and B-lymphocytes interact through idiotype–anti-idiotype reactions producing either stimulation or suppression. T–T interactions could occur through direct recognition of one T-cell receptor (TCR) by the other, or more usually by recognition of a processed TCR peptide associated with MHC. One of the anti-idiotype sets, $Ab_2\beta$, may bear an idiotype of similar shape to (i.e. provides an **internal image** of) the antigen. The same idiotype (●) may be shared by receptors of different specificity, the nonspecific parallel set (since the several hypervariable regions provide a number of potential idiotypic determinants and a given idiotype does not always form part of the epitope-binding site, i.e. the paratope), so that the anti-(anti-Id_1) does not necessarily bind the original antigen. (The following abbreviations are often employed: α as a prefix = anti; Id = idiotype; Ab_1 = Id; $Ab_2\alpha$ = αId not involving the paratope; $Ab_2\beta$ = internal image αId involving the paratope; Ab_3 = α(αId).)

derived from a TCR CDR3-associated idiotype presented by MHC class I.

A network is evident in early life

If the spleens of fetal mice which are just beginning to secrete immunoglobulin are used to produce hybridomas, an unusually high proportion are interrelated as idiotype–anti-idiotype pairs. This high level of idiotype connectivity is not seen in later life and suggests that these early cells, largely the $CD5^+$ **B-1 subset** (cf. p. 236), are programed to synthesize germ-line gene specificities which have network relationships.

Preoccupation of networks with self

Paradoxically, there is increasing evidence that preformed idiotype networks have what might seem at first sight a somewhat unhealthy interest in self-antigens. The CD5 IgM hybridomas produced from fetal mouse spleen with high idiotype connectivity have specificities similar to those of the '**natural antibodies**' which appear spontaneously in germ-free animals not exposed to exogenous antigens. Not only are many of them directed to polysaccharide antigens of common pathogens, e.g. phosphorylcholine, but a number of them have low affinity for self-components such as DNA, IgG, heat-shock proteins and cytoskeletal elements. The concept of a largely CD5 B-1 cell population forming an inward-looking world in which the component cells recognize and stimulate each other ceaselessly through their idiotypic receptor interactions to produce a range of IgM antibodies which provide an early defense against infection is most plausible. But the self-reactivity of many of these cells is enigmatic. Preset regulatory T-cell networks may also involve certain dominant autoantigens, such as heat-shock protein hsp65 and myelin basic protein from nervous tissue. Is recognition of self as important as nonself?

Irun Cohen has proposed the intriguing notion of an **immunological homunculus** (little man) in which a functional picture of the body is encoded within the immune system by regulatory network committees of

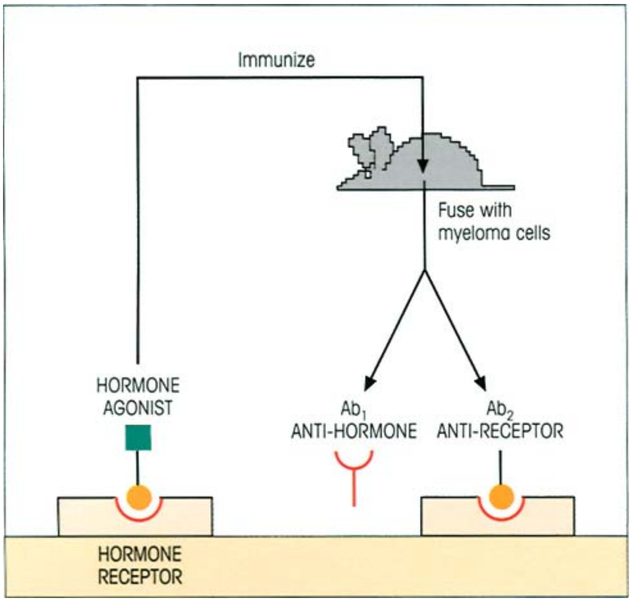

Figure 11.11. The spontaneous production of autoanti-idiotype during immunization with a hormone agonist. Hybridomas obtained from the immunized mouse secrete anti-hormone and anti-idiotype which reacts with the hormone receptor; this indicates a close relationship between hormone, receptor, anti-hormone and anti-idiotype, implying some connection between autoantibodies within the idiotype network and hormone systems.

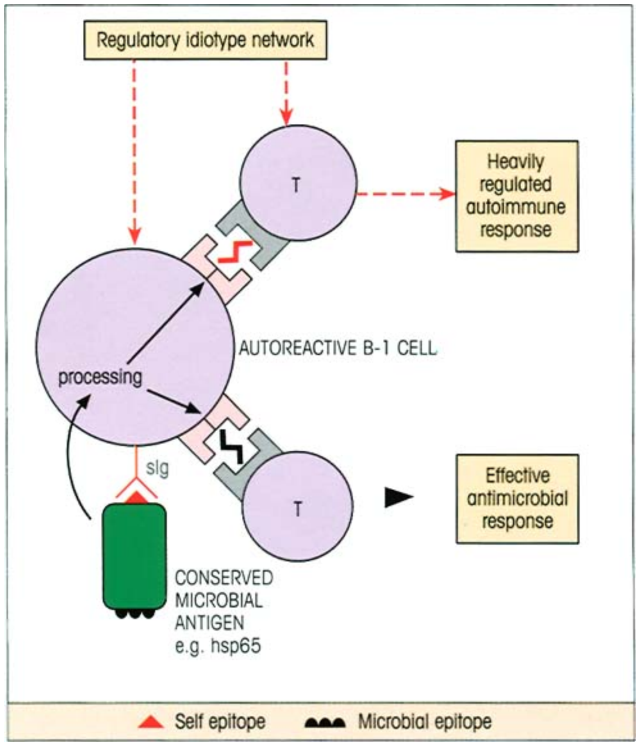

Figure 11.12. Speculation on the role of self-reacting CD5 B-1 cells in microbial infection. The surface immunoglobulin (sIg) receptor selectively captures the cross-reacting conserved bacterial antigen (in this case, heat-shock protein hsp65) and, after processing, presents self and microbe-specific epitopes associated with MHC class II. The autoreactive T-cell is heavily controlled by the idiotype regulatory network since (*ex hypothesi*) important self-antigens are dominantly encoded within the 'immunological homunculus', but the antimicrobial T-cell is free to mount a vigorous response. This accounts for the dominance of conserved antigens. (Based upon Cohen I.R. & Young D.B. (1991) *Immunology Today* **12**, 105.)

B- and T-cells which recognize certain dominant self-antigens representing the major organs in the body. (The analogy is with the neurological homunculus, a functional picture of the body in which the space occupied by a given neural network is directly related to the neurological importance of the organ it encodes, e.g. human visual and speech organs and canine olfactory organs are prominently represented.) The relevance of these ideas to the control of autoimmunity will be discussed in Chapter 19, but here we can speculate on how they might also relate to the response to infection. Consider a dominant microbial antigen such as hsp65 which is highly conserved in nature and bears several potential epitopes identical with the self-homolog. In an infection, any B-1 cells which recognize *self*-hsp65 will selectively focus the *bacterial* hsp65 onto their surface receptors (making it dominant over other bacterial antigens) and process it. The self-epitopes will be recognized by autoreactive T-cells which are highly regulated within the homunculus, whereas T-cells specific for the nonself hsp epitopes are not so constrained and will generate an effective immune response (figure 11.12).

Idiotypic regulation of immune responses

There have been a series of investigations based on the following cascade protocol. Antigen is injected into animal$_1$ and the antibody produced, Ab_1 (idiotype), is purified and injected into animal$_2$. Ab_2 (anti-idiotype) so formed is purified and used to immunize animal$_3$ and so on (figure 11.13). Consistently, it is found that Ab_2 (anti-Id_1) recognizes an idiotype (Id_1) on Ab_1 which is also strongly present in Ab_3. Ab_4 behaves like Ab_2 in seeing the common idiotype on Ab_1 and Ab_3. Nonetheless, although Ab_1 and Ab_3 share idiotypes, only a small fraction of Ab_3 reacts with the original antigen. This is the result one would expect if the idiotype was a cross-reacting Id (public Id) present on a variety of antibodies (and thus B-cell receptors) of different specificities. The anti-Id_1 (Ab_2), when injected into animal$_3$, would react with all B-cells bearing Id_1 (figure 11.13) and presumably trigger them to produce Id_1 antibodies, only a fraction of which have specificity for the original antigen.

Such frequently occurring and usually germ-line-

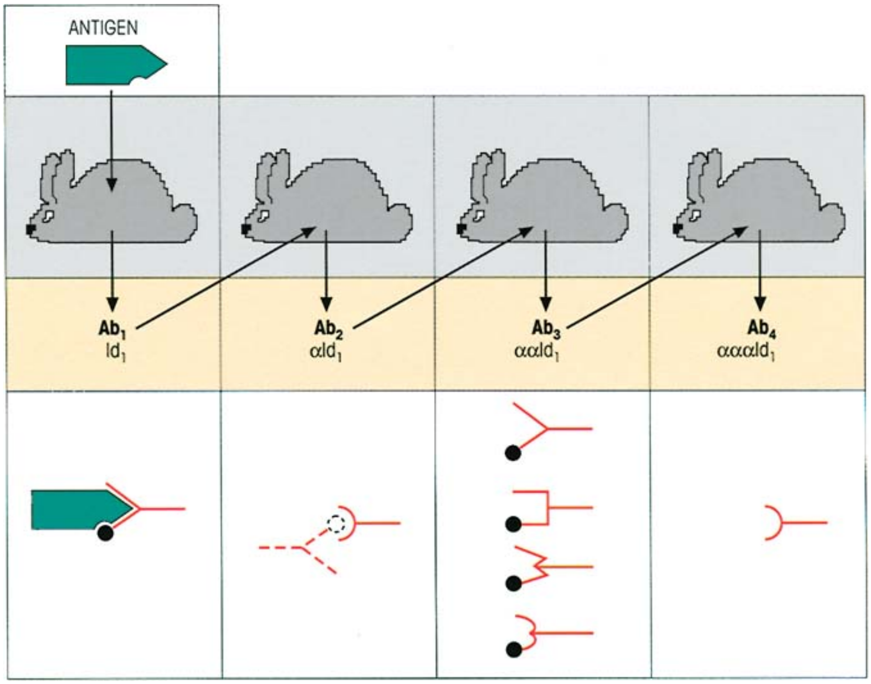

Figure 11.13. An idiotype cascade. Ab_1 produced by the antigen is injected into a second animal to produce Ab_2; this in turn is purified and injected into $animal_3$ and so on. Ab_2 and Ab_4 both react with an idiotype (●) on Ab_1 and Ab_3, but only a fraction of Ab_3 reacts with the original antigen. Bona and Paul (*Immunology Today* 1982, **3**, 230) interpret the results in terms of a common regulatory idiotype Id_1 shared by many antibodies other than those reacting with the original antigen, but recruited by the injection of anti-Id_1 (Ab_2) which stimulates the range of lymphocytes whose receptors bear this common or cross-reacting idiotype. On this basis, one can understand the paradoxical finding of Oudin and Cazenave that not all the Ig molecules bearing a given Id in response to an antigen can function as specific antibody since they belong to the nonspecific parallel set. The presence of large amounts of Id_1 in Ab_3 also suggests that the linear relationship through the cross-reacting Id_1 is dominant, with relatively insignificant branching through the variety of 'private' idiotypes on Ab_2 molecules (cf. figure 11.10) because of the low frequency of such idiotypes and their anti-idiotypes.

encoded idiotypes seem to be provoked fairly readily with anti-Id and are therefore candidates for **regulatory Id** which can be under some degree of control by a limited idiotypic network. Germane to this idea are the observations that, late in immunization, antibodies with utterly distinct specificities, directed against totally different epitopes on the same antigen, often bear a common or cross-reacting idiotype. Presumably, the first clone of B-cells to be expanded which bears a dominant cross-reacting Id can generate a population of regulatory Th cells which recognize this Id as well as antigen. Processing of internalized Ig receptor plus antigen leads to the expression of peptides derived from the idiotype and antigen in association with MHC class II; these B-cells can then access the full repertoire of antigen-specific and idiotype-specific Th cells. The latter may be of two types, one recognizing the native receptor idiotype (non-MHC-restricted) and the other, processed idiotype (MHC-restricted). From the complex mixture of B-lymphocytes activated by the other epitopes on the antigen, these Th cells will selectively recruit those with Id-positive receptors. We can now see how the antigen- and idiotype-specific Th synergize in the antibody response, the latter expanding Id-positive clones induced by the former.

The phenomenon of '**original antigenic sin**' occurs when the immune response becomes 'locked in' to particular epitopes originally encountered on a microorganism, such that it largely ignores even normally immunodominant epitopes during a subsequent encounter with an antigenically related but nonidentical microorganism. Although competition for antigen by the expanded population which forms the memory B-cells plays a major role, idiotype-specific memory Th cells could also contribute to this phenomenon.

Idiotype networks may also allow the immune response to 'tick over' for extended periods and maintain the memory cell population, while the presence of primed Th cells directed against a common Id on the various memory B-cells specific for a given antigen would increase their rate of mobilization during a secondary response.

Manipulation of the immune response through idiotypes

Quite low doses of anti-idiotype, of the order of nanograms, can greatly enhance the expression of the idiotype in the response to a given antigen, whereas doses in the microgram range lead to a suppression (figure 11.14). Thus the idiotypic network provides interesting opportunities to manipulate the immune response, particularly in hypersensitivity states such as autoimmune disease, allergy and graft rejection. However, the B-cell response is normally so diverse, suppression by anti-Id is likely to prove difficult; even when the response is dominated by a public Id and that Id is suppressed, compensatory expansion of clones bearing other idiotypes ensures that the fall in the total antibody titer is relatively undramatic (cf. figure 11.14). Conceivably, Th cells may express a narrower spectrum of idiotypes, thereby being more susceptible to suppression by Id autoimmunization. Reports that 'vaccination' with irradiated lines of Th cells specific for brain or thyroid antigens prevents the induction of experimental autoimmunity against the relevant organ are encouraging. A totally different approach would be to use monoclonal anti-Id of the 'antigen internal image' set (figure 11.10) to stimulate antigen-specific T-suppressors capable of turning off B-cells directed to other epitopes on the antigen through bridging by the antigen itself (cf. figure 11.8).

Since we know that under suitable conditions anti-Id can also stimulate antibody production, it might be possible to use 'internal image' monoclonal anti-Ids as 'surrogate' antigens for immunization in cases where the antigen is difficult to obtain in bulk—for example, antigens from parasites such as filaria or the weak embryonic antigens associated with some cancers. Another example is where protein antigens obtained by chemical synthesis or gene cloning fail to fold into the configuration of the native molecule; this is not a problem with the anti-Id which by definition has been selected to have the shape of the antigenic epitope.

At this stage, if the reader is feeling a little groggy, try a glance at figure 11.15 which attempts to summarize the main factors currently thought to modulate the immune response.

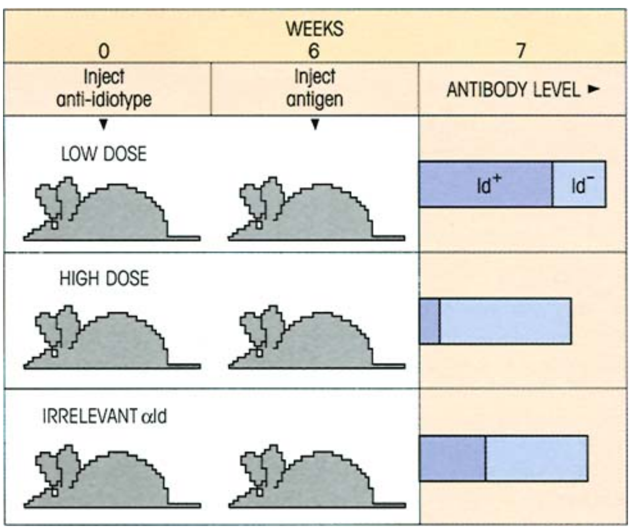

Figure 11.14. Modulation of a major idiotype in the antibody response to antigen by anti-idiotype. In the example chosen, the idiotype is present in a substantial proportion of the antibodies produced in controls injected with irrelevant anti-Id plus antigen (i.e. this is a public or cross-reacting Id; see p. 44). Pretreatment with 10 ng of a monoclonal anti-Id greatly expands the Id^+ antibody population, whereas prior injection of 10 μg of anti-Id almost completely suppresses expression of the idiotype without having any substantial effect on total antibody production due to a compensatory increase in Id^- antibody clones.

THE INFLUENCE OF GENETIC FACTORS

Some genes affect general responsiveness

Mice can be selectively bred for high or low antibody responses through several generations to yield two lines, one of which consistently produces high-titer antibodies to a variety of antigens, and the other, antibodies of relatively low titer (figure 11.16; Biozzi and colleagues). Out of the ten or so different genetic loci involved, some give rise to a higher rate of B-cell proliferation and differentiation, while one or more affect macrophage behavior.

Antigen receptor genes are linked to the immune response

Clearly, the Ig and TCR *V*, *D* and *J* genes encoding the specific recognition sites of the lymphocyte antigen receptors are of fundamental importance to the acquired immune response. However, since the mechanisms for generating receptor diversity from the available genes are so powerful (cf. p. 65), immunodeficiency is unlikely to occur as a consequence of a poor Ig or TCR variable region gene repertoire. Nevertheless, just occasionally, we see holes in the repertoire due to the absence of a gene; failure to respond to the sugar polymer α1–6 dextran is a feature of animals without a particular immunoglobulin *V* gene, and mice lacking the $V\alpha_2$ TCR gene cannot mount a cytotoxic T-cell response to the male H-Y antigen.

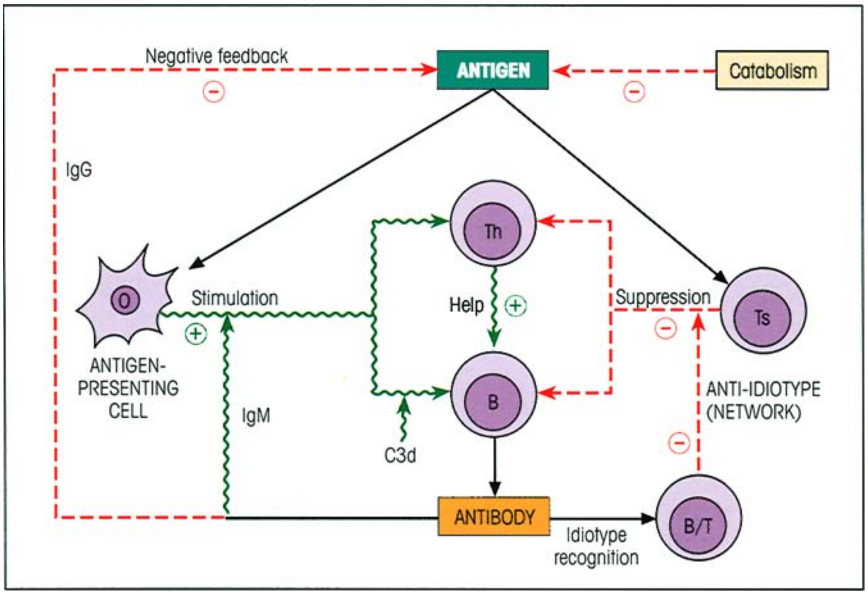

Figure 11.15. Regulation of the immune response. Th, T-helper cell; Ts, T-suppressor cell. T-help for cell-mediated immunity will be subject to similar regulation. Some of these mechanisms may be interdependent; for example, one could envisage anti-idiotypic antibody acting in concert with a suppressor T-cell by binding to its Fc receptor, or suppressor T-cells with specificity for the idiotype on Th or B-cells. To avoid too many confusing arrows, we have omitted the recruitment of B-cells by anti-idiotypic Th cells and direct activation of anti-idiotype Ts by idiotype-positive Th cells.

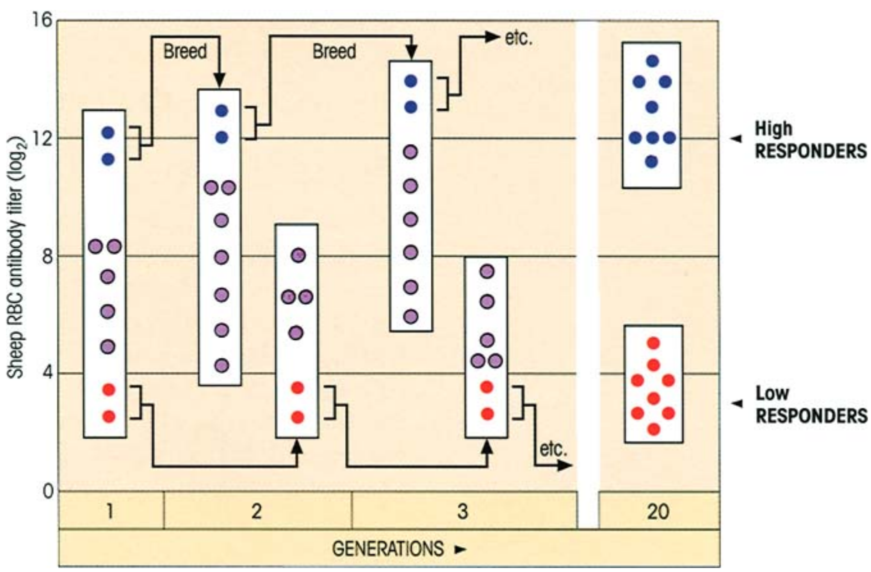

Figure 11.16. Selective breeding of high and low antibody responders (after Biozzi and colleagues). A foundation population of wild mice (with crazy mixed-up genes and great variability in antibody response) is immunized with sheep red blood cells (SRBC), a multideterminant antigen. The antibody titer of each individual mouse is shown by a circle. The male and female giving the highest titer antibodies (●) were bred and their litter challenged with antigen. Again, the best responders were bred together and so on for 20 generations when all mice were high responders to SRBC and a variety of other antigens. The same was done for the poorest responders (●), yielding a strain of low responder animals. The two lines are comparable in their ability to clear carbon particles or sheep erythrocytes from the blood by phagocytosis, but macrophages from the high responders present antigen more efficiently (cf. p. 158). On the other hand, the low responders survive infection by *Salmonella typhimurium* better and their macrophages support much slower replication of *Listeria* (cf. p. 262), indicative of an inherently more aggressive microbicidal ability.

Immune response can be influenced by the MHC

There was much excitement when it was first discovered that the antibody responses to a number of thymus-dependent antigenically simple substances are determined by genes mapping to the MHC. For example, mice of the *H-2*^{b} haplotype respond well to the synthetic branched polypeptide (T,G)-A–L, whereas *H-2*^{k} mice respond poorly (table 11.2).

It was said that mice of the *H-2*^{b} haplotype (i.e. a particular set of *H-2* genes) are **high responders** to (T,G)-A–L because they possess the appropriate immune response (*Ir*) gene. With another synthetic antigen, (H,G)-A–L, having histidine in place of tyrosine, the

Table 11.2. H-2 haplotype linked to high, low and intermediate immune responses to synthetic peptides. (T,G)-A–L, polylysine with polyalanine side-chains randomly tipped with tyrosine and glutamine; (H,G)-A–L, the same with histidine in place of tyrosine.

ANTIGEN	H-2 HAPLOTYPE				
	b	k	d	a	s
(T,G)-A–L	Hi	Lo	Int	Lo	Lo
(H,G)-A–L	Lo	Hi	Int	Hi	Lo

Table 11.3. Mapping of the *Ir* gene for (H,G)-A–L responses by analysis of different recombinant strains.

Strain	H-2 region				(H,G)-A–L Response
	K	A	E	D	
A	k	k	k	b	Hi
A.TL	s	k	k	b	Hi
B.IO.A (4R)	k	k	b	b	Hi
B.IO	b	b	b	b	Lo
A.SW	s	s	s	s	Lo

position is reversed, the 'poor (T,G)-A–L responders' now giving a good antibody response and the 'good (T,G)-A–L responders' a weak one, showing that the capacity of a particular strain to give a high or low response varies with the individual antigen (table 11.2). These relationships are only apparent when antigens of highly restricted structure are studied because the response to each single determinant is controlled by an *Ir* gene and it is less likely that the different determinants on a complex antigen will all be associated with consistently high or consistently low responder *Ir* genes; however, although one would expect an average of randomly high and low responder genes, since the various determinants on most thymus-dependent complex antigens are structurally unrelated, the outcome will be biased by the dominance of one or more epitopes (cf. p. 201). Thus H-2-linked immune responses have been observed not only with relatively simple polypeptides, but also with transplantation antigens from another strain and autoantigens where merely one or two determinants are recognized as foreign by the host. With complex antigens, in most but not all cases, H-2 linkage is usually only seen when the dose administered is so low that just one immunodominant determinant is recognized by the immune system. In this way, reactions controlled by *Ir* genes are distinct from the overall responsiveness to a variety of complex antigens which is a feature of the Biozzi mice (above).

The Ir genes map to the H-2I region and control T–B cooperation

Table 11.3 gives some idea of the type of analysis used to map the *Ir* genes. The three high responder strains have individual *H-2* genes derived from prototypic pure strains which have been interbred to produce recombinations within the H-2 region. The only genes they have in common are A^k and D^b; since the B.10 strain bearing the D^b gene is a low responder, high response must be linked in this case to possession of A^k. The I region molecules must represent the *Ir* gene product since a point mutation in the H-2A subregion in one strain led to a change in the class II molecule at a site affecting its polymorphic specificity and changed the mice from high to low responder status with respect to their thymus-dependent antibody response to antigen *in vivo*. The mutation also greatly reduced the proliferation of T-cells from immunized animals when challenged *in vitro* with antigen plus appropriate presenting cells, and there is a good correlation between antigen-specific T-cell proliferation and the responder status of the host. The implication that **responder status may be linked to the generation of Th cells** is amply borne out by adoptive transfer studies showing that irradiated (H-2^b×H-2^k) F1 mice make good antibody responses to (T,G)-A–L when reconstituted with antigen-primed B-cells from another F1 plus T-cells from a primed H-2^b (high responder); T-cells from the low responder H-2^k mice only gave poor help for antibody responses. This also explains why these *H-2* gene effects are seen with thymus-dependent but not T-independent antigens.

Three mechanisms have been proposed to account for class II-linked high and low responsiveness.

1 *Defective presentation*. In a high responder, processing of antigen and its recognition by a corresponding T-cell lead to lymphocyte triggering and clonal expansion (figure 11.17a). Although there is (and has to be) considerable degeneracy in the specificity of the class II groove for peptide binding, variation in certain key residues can alter the strength of binding to a particular peptide (cf. p. 98) and convert a high to a low responder because the MHC fails to present antigen to the reactive T-cell (figure 11.17b). Sometimes the natural processing of an antigen in a given individual does not produce a peptide which fits well into their MHC molecules. One study showed that a cytotoxic T-cell clone restricted to HLA-A2, which recognized residues 58–68 of influenza A virus matrix protein, could cross-react with cells from an A69 subject pulsed with the same peptide; nonetheless, the clone failed to

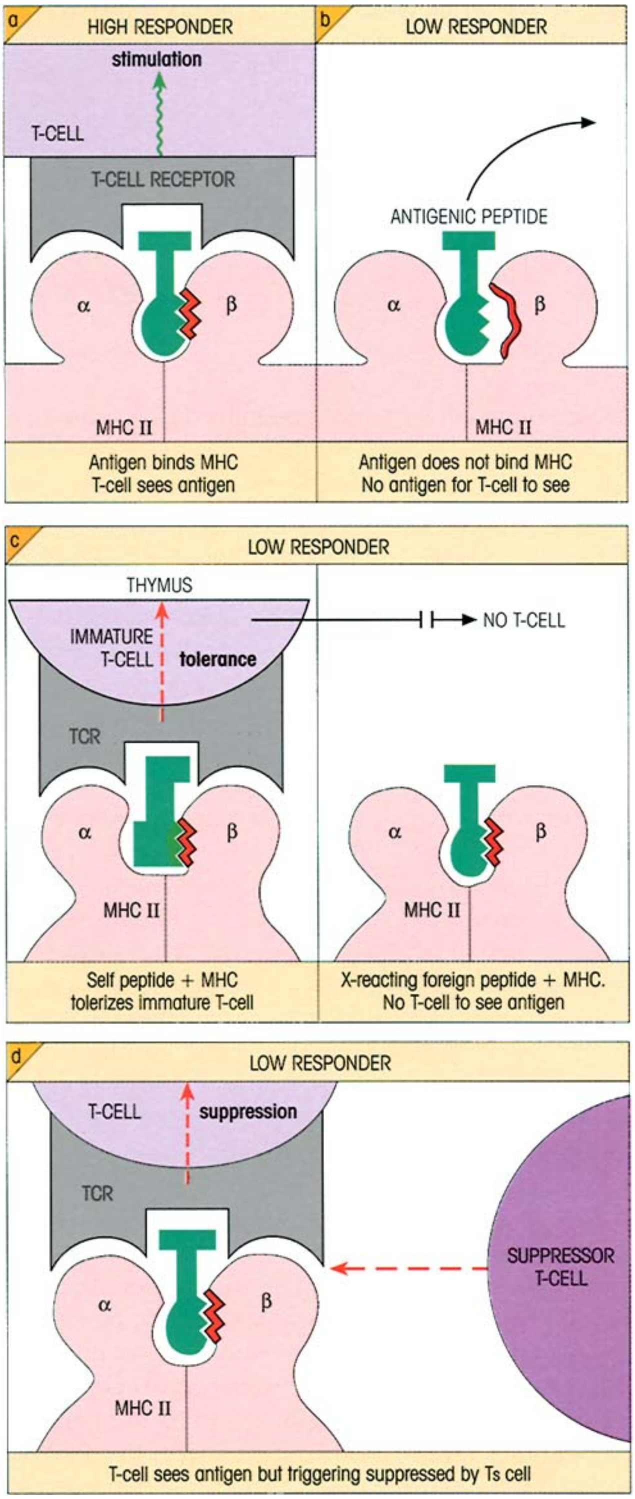

Figure 11.17. Different mechanisms can account for low T-cell response to antigen in association with MHC class II.

recognize A69 cells *infected* with influenza A virus. Interestingly, individuals with the HLA-A69 class I MHC develop immunity to a different epitope on the same protein.

2 *Defective T-cell repertoire*. T-cells with moderate to high affinity for self-MHC molecules and their complexes with processed self-antigens will be rendered unresponsive (cf. tolerance induction, p. 229), so creating a 'hole' in the T-cell repertoire. If there is a cross-reaction, i.e. similarity in shape at the T-cell recognition level between a foreign antigen and a self-molecule which has already induced unresponsiveness, the host will lack T-cells specific for the foreign antigen and therefore be a low responder (figure 11.17c). To take a concrete example, mice of DBA/2 strain respond well to the synthetic peptide polyglutamyl, polytyrosine (GT), whereas BALB/c mice do not, although both have identical class II genes. BALB/c B-cell blasts express a structure which mimics GT and the presumption would be that self-tolerance makes these mice unresponsive to GT. This was confirmed by showing that DBA/2 mice made tolerant by a small number of BALB/c hematopoietic cells were changed from high to low responder status. To round off the story in a very satisfying way, DBA/2 mice injected with BALB/c B-blasts, induced by the polyclonal activator lipopolysaccharide, were found to be primed for GT.

3 *T-suppression*. We would like to refer again to the MHC-restricted low responsiveness which can occur to relatively complex antigens (see p. 205), since it illustrates the notion that low responder status can arise as an expression of CD8 T-suppressor activity (figure 11.17d). Low response can be dominant in class II heterozygotes, indicating that suppression can act against Th restricted to any other class II molecule. In this it differs from models 1 and 2 above where high response is dominant in a heterozygote because the factors associated with the low responder gene cannot influence the activity of the high responder. Overall, it seems likely that each of the three models may provide the basis for class II-linked *Ir* gene phenomena in different circumstances.

Factors influencing the genetic control of the immune response are summarized in figure 11.18.

REGULATORY IMMUNONEUROENDOCRINE NETWORKS

There is a danger, as one focuses more and more on the antics of the immune system, of looking at the body as a collection of myeloid and lymphoid cells roaming around in a big sack and of having no regard to the integrated physiology of the organism. Within the wider physiological context, attention has been drawn increasingly to interactions between immunological and neuroendocrine systems.

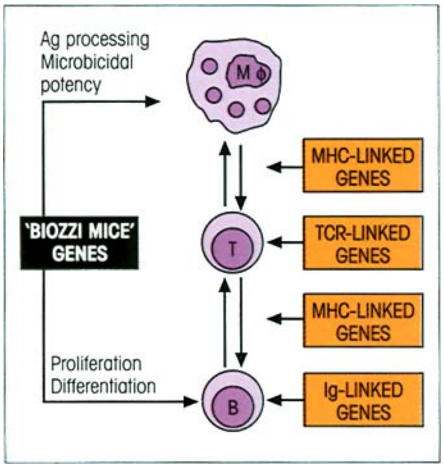

Figure 11.18. Genetic control of the immune response.

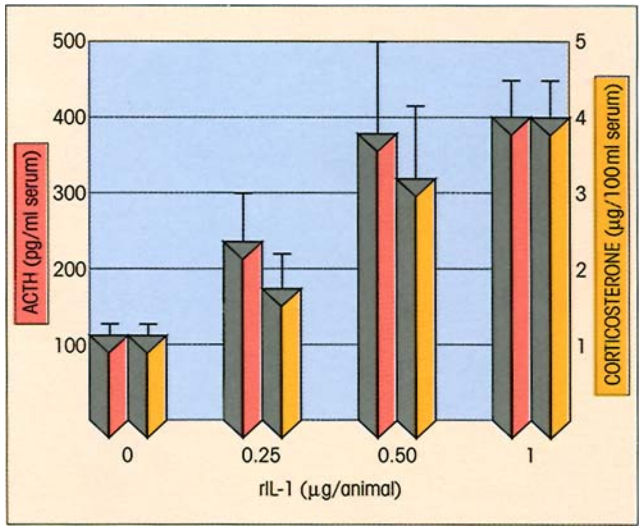

Figure 11.19. Enhancement of ACTH and corticosterone blood levels in C3H/HeJ mice 2 hours after injection of recombinant IL-1 (values are means ± SEM for groups of seven or eight mice). The significance of the mouse strain used is that it lacks receptors for bacterial lipopolysaccharide (LPS), and so the effects cannot be attributed to LPS contamination of the IL-1 preparation. (Reprinted from Besedovsky H., del Rey A., Sorkin E. & Dinarello C.A. (1986) *Science* **233**, 652, with permission. Copyright © 1986 by the AAAS.)

Immunological cells have the receptors which enable them to receive signals from a whole range of hormones: corticosteroids, insulin, growth hormone, estradiol, testosterone, prolactin, β-adrenergic agents, acetylcholine, endorphins and enkephalins. By and large, glucocorticoids and androgens depress immune responses, whereas estrogens, growth hormone, thyroxine and insulin do the opposite.

A neuroendocrine feedback loop affecting immune responses

The secretion of **glucocorticoids** is a major response to stresses induced by a wide range of stimuli, such as extreme changes of temperature, fear, hunger and physical injury. They are also released as a consequence of immune responses and limit those responses in a neuroendocrine feedback loop. Thus, IL-1 (figure 11.19), IL-6 and TNF are capable of stimulating glucocorticoid synthesis and do so through the hypothalamic–pituitary–adrenal axis. This, in turn, leads to the downregulation of Th1 and macrophage activity, so completing the negative feedback circuit (figure 11.20). However, the glucocorticoid dexamethasone can prevent *a*ctivation-*i*nduced *c*ell *d*eath (AICD) in T-cells by inducing expression of GILZ (*g*lucocorticoid-*i*nduced *l*eucine *z*ipper). The situation is therefore somewhat complex because glucocorticoids can themselves trigger apoptosis in T-cells, yet counteract apoptosis activated by peptide–MHC interaction with the TCR. In the absence of glucocorticoid, the activation of T-cells by peptide–MHC leads to a progressive loss of GILZ and eventual cell death by apoptosis. In contrast, if activation through the TCR occurs in the presence of glucocorticoids, then expression of GILZ is increased and the cells are protected from AICD, perhaps via an effect on NFκB.

It has been shown that adrenalectomy prevents spontaneous recovery from *e*xperimental *a*llergic *e*ncephalomyelitis (EAE), a demyelinating disease with progressive paralysis produced by myelin basic protein in complete Freund's adjuvant which, however, can be blocked by implants of corticosterone. Spontaneous recovery from EAE in intact animals is associated with a dominance of Th2 autoantigen-specific clones, indicative of the view that glucocorticoids suppress Th1 and may augment Th2 cells. Individuals with a genetic predisposition to high levels of stress-induced glucocorticoids would be expected to have increased susceptibility to infections with intracellular pathogens such as *Mycobacterium leprae* which require effective Th1 cell-mediated immunity for their eradication.

Recent experiments have demonstrated that neonatal exposure to bacterial endotoxin (LPS) not only exerts a long-term influence on endocrine and central nervous system development, but substantially affects predisposition to inflammatory disease and therefore appears to program or 'reset' the functional development of both the endocrine and immune systems. Thus, in adult life, rats which had been exposed to endotoxin during the first week of life had higher basal levels of corticosterone compared with control ani-

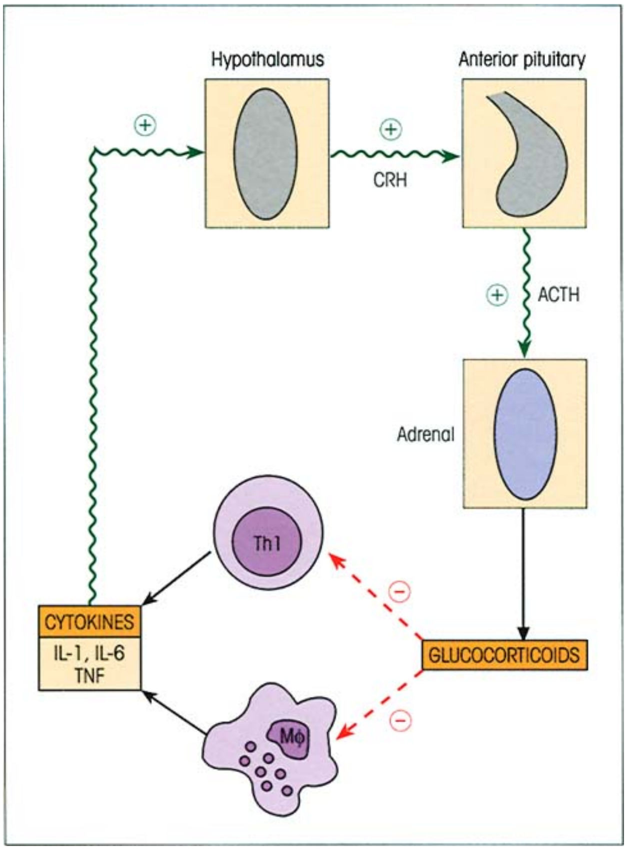

Figure 11.20. Glucocorticoid negative feedback on cytokine production. CRH, corticotropin-releasing hormone; ACTH, adrenocorticotropic hormone. Evidence for another type of circuit encompassing hormone receptor, hormone, anti-hormone and internal image anti-idiotype was presented in figure 11.11 and may be of relevance to the pathogenesis of autoimmune disorders directed against hormone receptors (see p. 422). More generally, cells of the immune system in both primary and secondary lymphoid organs can produce hormones and neuropeptides, while classical endocrine glands as well as neurons and glial cells can synthesize cytokines and appropriate receptors. Production of prolactin and its receptors by peripheral lymphoid cells and thymocytes is worthy of attention. Lymphocyte expression of the prolactin receptor is upregulated following activation and, in autoimmune disease, witness the beneficial effects of bromocriptine, an inhibitor of prolactin synthesis, in the NZB × W model of murine SLE (cf. p. 403).

mals, and showed a greater increase in corticosterone levels in response to noise stress and a more rapid rise in corticosterone levels following challenge with LPS.

Sex hormones come into the picture

Estrogen is said to be the major factor influencing the more active immune responses in females relative to males. They have higher serum Ig and secreted IgA levels, a higher antibody response to T-independent antigens, relative resistance to T-cell tolerance and greater resistance to infections. Females are also far more susceptible to autoimmune disease, an issue that will be discussed in greater depth in Chapter 19, but here let us note that oral contraceptives can induce flares of the autoimmune disorder *s*ystemic *l*upus *e*rythematosus (SLE; see p. 425) and the mildly androgenic adrenal hormone, **dehydroepiandrosterone (DHEA)**, can significantly prolong the lifespan of (NZB×W) F1 females with the murine model of SLE. DHEA has positive effects on Th1 at the expense of Th2 cells and can clearly antagonize the inhibitory effects of cortisol on thymocytes and T-cell proliferation. As they say in the tabloids, watch this space.

'Psychoimmunology'

The thymus, spleen and lymph nodes are richly innervated by the sympathetic nervous system. The enzyme dopamine β-hydroxylase catalyses the conversion of dopamine to the catecholamine neurotransmitter norepinephrine which is released from sympathetic neurons in these tissues. Mice in which the gene for this enzyme has been deleted by homologous recombination exhibited enhanced susceptibility to infection with the intracellular pathogen *Mycobacterium tuberculosis* and impaired production of the Th1 cytokines IFNγ and TNF in response to the infection. Although these animals showed no obvious developmental defects in their immune system, impaired Th1 responses were also found following immunization of these mice with the hapten TNP coupled to KLH. These observations suggest that norepinephrine can play a role in determining the potency of the immune response.

Denervated skin shows greatly reduced leukocyte infiltration in response to local damage, implicating cutaneous neurons in the recruitment of leukocytes. Sympathetic nerves which innervate lymphatic vessels and lymph nodes are involved in regulating the flow of lymph and may participate in controlling the migration of β-adrenergic receptor-bearing dendritic cells from inflammatory sites to the local lymph nodes. Mast cells and nerves often have an intimate anatomical relationship and nerve growth factor causes mast cell degranulation. The gastrointestinal tract also has extensive innervation and a high number of immune effector cells. In this context, the ability of substance P to stimulate, and of somatostatin to inhibit, proliferation of Peyer's patch lymphocytes may prove to have more than a trivial significance. The pituitary hormone prolactin has also been brought to our attention by the experimental observation that inhibition of prolactin secretion by bromocriptine suppresses Th activity.

There seems to be an interaction between inflammation and nerve growth in regions of wound healing and repair. Mast cells are often abundant, IL-6 induces neurite growth and IL-1 enhances production of nerve growth factor in sciatic nerve explants. IL-1 also increases slow-wave sleep when introduced into the lateral ventricle of the brain, and both IL-1 and interferon produce pyrogenic effects through their action on the temperature-controlling center.

Although it is not clear just how these diverse neuroendocrine effects fit into the regulation of immune responses, at a more physiological level, stress and circadian rhythms modify the functioning of the immune system. Factors such as restraint, noise and exam anxiety have been observed to influence a number of immune functions including phagocytosis, lymphocyte proliferation, NK activity and IgA secretion. Amazingly, the delayed-type hypersensitivity Mantoux reaction in the skin can be modified by hypnosis. An elegant demonstration of nervous system control is provided by studies showing suppression of conventional immune responses and enhancement of NK cell activity by Pavlovian conditioning. In the classic Pavlovian paradigm, a stimulus such as food that unconditionally elicits a particular response, in this case salivation, is repeatedly paired with a neutral stimulus that does not elicit the same response. Eventually, the neutral stimulus becomes a conditional stimulus and will elicit salivation in the absence of food. Rats were given cyclophosphamide as a unconditional and saccharin as a conditional stimulus repeatedly; subsequently, there was a depressed antibody response when the animals were challenged with antigen together with just the conditional stimulus, saccharin. As more and more data accumulate, it is becoming clearer how immunoneuroendocrine networks could play a role in allergy and in autoimmune diseases such as rheumatoid arthritis, insulin-dependent diabetes mellitus and multiple sclerosis.

EFFECTS OF DIET, EXERCISE, TRAUMA AND AGE ON IMMUNITY

Malnutrition diminishes the effectiveness of the immune response

The greatly increased susceptibility of undernourished individuals to infection can be attributed to many factors: poor sanitation and personal hygiene, overcrowding and inadequate health education. But, in addition, there are gross effects of **protein-calorie malnutrition** on immunocompetence. The widespread atrophy of lymphoid tissues and the 50% reduction in circulating CD4 T-cells underlie **serious impairment of cell-mediated immunity**. Antibody responses may be intact but they are of lower affinity; phagocytosis of bacteria is relatively normal but the subsequent intracellular destruction is defective.

Deficiencies in pyridoxine, folic acid and vitamins A, C and E result in generally impaired immune responses. **Vitamin D is an important regulator**. It is produced not only by the UV-irradiated dermis, but also by activated macrophages, the hypercalcemia associated with sarcoidosis being attributable to production of the vitamin by macrophages in the active granulomas. The vitamin is a potent inhibitor of T-cell proliferation and of Th1 cytokine production. This generates a neat feedback loop at sites of inflammation where macrophages activated by IFNγ produce vitamin D which suppresses the T-cells making the interferon. It also downregulates antigen presentation by macrophages and promotes multinucleated giant cell formation in chronic granulomatous lesions. Nonetheless, as a further emphasis of the potential duality of the CD4 helper subsets, it promotes Th2 activity, especially at mucosal surfaces: quite a busy little vitamin. Zinc deficiency is rather interesting; this greatly affects the biological activity of thymus hormones and has a major effect on cell-mediated immunity, perhaps as a result. Iron deficiency impairs the oxidative burst in neutrophils since the flavocytochrome NADP oxidase is an iron-containing enzyme.

Of course there is another side to all this in that moderate restriction of total calorie intake and/or marked reduction in fat intake ameliorates age-related diseases such as autoimmunity. Oils with an n-3 double bond, such as fish oils, are also protective, perhaps due to increased synthesis of immunosuppressive prostaglandins.

Given the overdue sensitivity to the importance of environmental contamination, it is important to monitor the nature and levels of pollution that may influence immunity. Here is just one example: polyhalogenated organic compounds (such as polychlorinated biphenyls) steadily pervade the environment and, being stable and lipophilic, accumulate readily in the aquatic food chain where they largely resist metabolic breakdown. It was shown that Baltic herrings with relatively high levels of these pollutants, as compared with uncontaminated Atlantic herrings, were immunotoxic when fed to captive harbor seals, suggesting one reason why seals along the coasts of northwestern Europe succumbed so alarmingly to infection with the otherwise nonvirulent phocine distemper virus in 1988.

Other factors

Exercise, particularly severe exercise, induces stress and raises plasma levels of cortisol, catecholamines, IFNα, IL-1, β-endorphin and metenkephalin. It can lead to reduced IgA levels, immune deficiency and increased susceptibility to infection. Maniacal joggers and other such like masochists—you have been warned!

Multiple traumatic injury, surgery and major burns are also immunosuppressive and so contribute to the increased risk of sepsis. Corticosteroids produced by stressful conditions, the immunosuppressive prostaglandin E_2 released from damaged tissues and bacterial endotoxin derived from the disturbance of gut flora are all factors which influence the outcome after trauma.

Accepting that the problem of understanding the mechanisms of **aging** is a tough nut to crack, it is a trifle disappointing that the easier task of establishing the influence of age on immunological phenomena is still not satisfactorily accomplished. Perhaps the elderly population is skewed towards individuals with effective immune systems which give a survival advantage. Be that as it may, IL-2 production by peripheral blood lymphocytes (figure 11.21) and T-cell-mediated functions such as delayed-type hypersensitivity reactions to common skin test antigens decline with age and so, it is thought, does T-suppression, although this is a notoriously elusive function to measure.

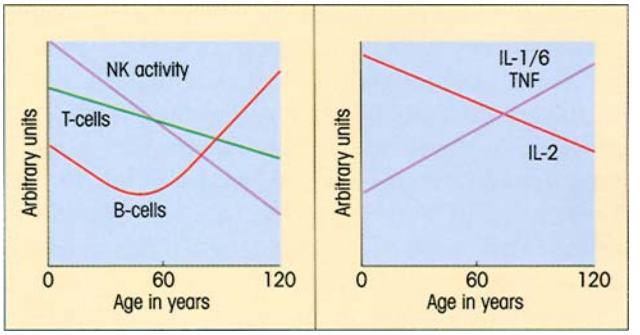

Figure 11.21. Age trends in some immunological parameters. (Based on Franceschi C., Monti D., Sansoni P. & Cossarizza A. (1995) *Immunology Today* **16**, 12.)

Most B-cell responses to exogenous antigens are not dramatically changed with the passage of time, but one of the few well-founded observations concerns the relative increase in agalactosyl oligosaccharides on the Cγ2 domains of IgG (cf. p. 430) in parallel with a rise in IL-6 in the older age groups (figure 11.21). This is accompanied by a decrease in DHEA, and it may be significant that injection of the androgen into elderly mice leads to a fall in circulating IL-6 concentrations. Do these studies provide us with a clue to the increased prevalence of autoantibodies in our senior citizens?

SUMMARY

Control by antigen

- Immune responses are largely antigen driven. As the level of antigen falls, so does the intensity of the response.
- Antigens can compete with each other: a result of competition between processed peptides for the available MHC grooves.

Feedback control by complement and antibody

- Early IgM antibodies and C3d boost antibody responses, whereas IgG inhibits responses via the Fcγ receptor on B-cells.

T-cell regulation

- Activated T-cells express members of the TNF receptor family, including Fas, which act as death receptors and restrain unlimited clonal expansion by a process referred to as activation-induced cell death (AICD).
- At high levels of antigen, T-cells which suppress T-helpers emerge presumably as feedback control of excessive Th expansion.
- Suppressor and helper epitopes on the same molecule can be discrete.
- In some instances of suppression, the effectors are CD8 T-cells restricted either directly (which would be strange) or indirectly through a CD4 intermediary.
- Suppression may well be due to T–T interaction on the surface of antigen-presenting cells. Just as Th1 and Th2 cells mutually inhibit each other through production of their respective cytokines IFNγ and IL-4/10, so there may be two types of CD8 cells with suppressor activity: one of Tc2 type found in lepromatous leprosy patients making IL-4 and suppressing Th1 cells, and the other Tc1 cells making IFNγ capable of suppressing Th2 cells.

(continued)

Idiotype networks

• Antigen-specific receptors on lymphocytes can interact with the idiotypes on the receptors of other lymphocytes to form a network (Jerne).
• Anti-idiotypes can be induced by autologous idiotypes.
• An idiotype network involving mostly CD5 B-1 cells is evident in early life.
• T-cell idiotypic interactions can also be demonstrated.
• Preset idiotypic networks involve a number of lymphocytes with self-reactivity for dominant autoantigens. It is speculated that this preoccupation with self helps to regulate unwanted autoimmune reactions and may help to target the response to infectious agents on dominant conserved antigens like hsp65 which cross-react with self.
• Idiotypes which occur frequently and are shared by a multiplicity of antibodies (public or cross-reacting Id) are targets for regulation by anti-idiotypes in the network, thus providing a further mechanism for control of the immune response.
• The network offers the potential for therapeutic intervention to manipulate immunity.

Genetic factors influence the immune response

• Approximately 10 genes control the overall antibody response to complex antigens: some affect macrophage antigen processing and microbicidal activity and some the rate of proliferation of differentiating B-cells.
• Immunoglobulin and TCR genes are very adaptable because they rearrange to create the antigen receptors, but 'holes' in the repertoire can occur.
• Immune response genes are located in the MHC class II locus and control the interactions required for T–B collaboration.
• Class II-linked high and low responsiveness may be due to defective presentation by MHC, a defective T-cell repertoire caused by tolerance to MHC + self-peptides and T-suppression.

Immunoneuroendocrine networks

• Immunological, neurological and endocrinological systems interact, forming regulatory circuits.
• Feedback by cytokines augments the production of corticosteroids and is important because this shuts down Th1 and macrophage activity.
• Estrogens may be largely responsible for the more active immune responses in females relative to males. The male hormone, DHEA, prolongs life in females with the murine equivalent of the human autoimmune disease SLE.

Effects of diet and other factors on immunity

• Protein-calorie malnutrition grossly impairs cell-mediated immunity and phagocyte microbicidal potency.
• Exercise, trauma, age and environmental pollution can all act to impair immune mechanisms. The pattern of cytokines produced by peripheral blood cells changes with age, IL-2 decreasing and TNF, IL-1 and IL-6 increasing; the latter is associated with a lowered DHEA level.

Factors influencing the bias between Th1 and Th2 subsets

• These have figured with some prominence in this chapter and a summary of some of the major influences on the balance between Th1 and Th2 responses is presented in figure 11.22.

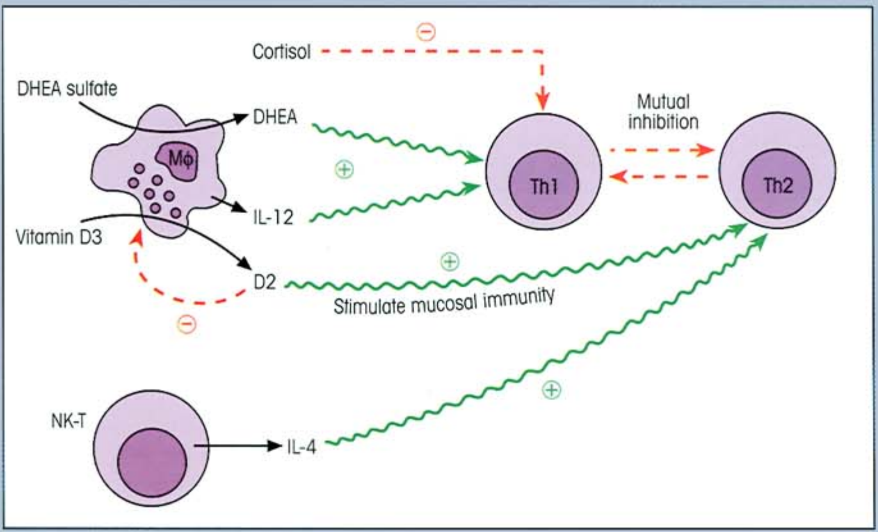

Figure 11.22. Summary of major factors affecting Th1/Th2 balance. Preferential stimulation of mucosal antibody synthesis by vitamin D involves the promotion of dendritic cell migration to the Peyer's patches. By downregulating macrophage activity, Th1 effectiveness is decreased. Cortisol and dehydroepiandrosterone (DHEA) are products of the adrenal and have opposing effects on the Th1 subset. A relative deficiency of DHEA will lead to poor Th1 performance. NK-T cells bear an $\alpha\beta$ TCR, the natural killer cell marker NK1.1, and secrete cytokines including IL-4 which stimulate Th2 cells.

FURTHER READING

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Ontogeny and phylogeny

INTRODUCTION

Hematopoiesis originates in the early yolk sac but, as embryogenesis proceeds, this function is taken over by the fetal liver and finally by the bone marrow where it continues throughout life. The **hematopoietic stem cell which gives rise to the formed elements of the blood** (figure 12.1) can be shown to be multipotent, to seed other organs and to have a relatively unlimited capacity to renew itself through the creation of further stem cells. Thus an animal can be completely protected against the lethal effects of high doses of irradiation by injection of bone marrow cells which will repopulate its lymphoid and myeloid systems. The capacity for self-renewal is not absolute and declines with age in parallel with a shortening of the telomeres and a reduction in telomerase, the enzyme which repairs the shortening of the ends of chromosomes which would otherwise occur at every round of cell division.

HEMATOPOIETIC STEM CELLS

We have come a long way towards the goal of isolating highly purified populations of hematopoietic stem cells, although not all agree that we have yet achieved it. In the mouse, the most likely candidate, at least a *very* early progenitor, is the cell with the following surface phenotype: high expression of MHC, low positivity for Thy-1, clearly positive for Sca-1, AA4.1 and c-kit, and for the adhesion molecule PGP-1, negative or only weakly positive for Lin, and negative for B220, Mac-1, Gr-1 and CD8 (the last four being markers for B-cells, macrophages, granulocytes and cytotoxic T-cells, respectively). Impressively, less than 100 of such cells can prevent death in a lethally irradiated animal. Recently, a 'molecular phenotype' of the mouse stem cell has been obtained by analysis of subtracted cDNA libraries from highly purified fetal liver stem cells. It was revealed that there is a very high degree of precision in the control of transcription at each stage of the hematopoietic hierarchy. Certain molecules are more predominant in either fetal or adult hematopoietic

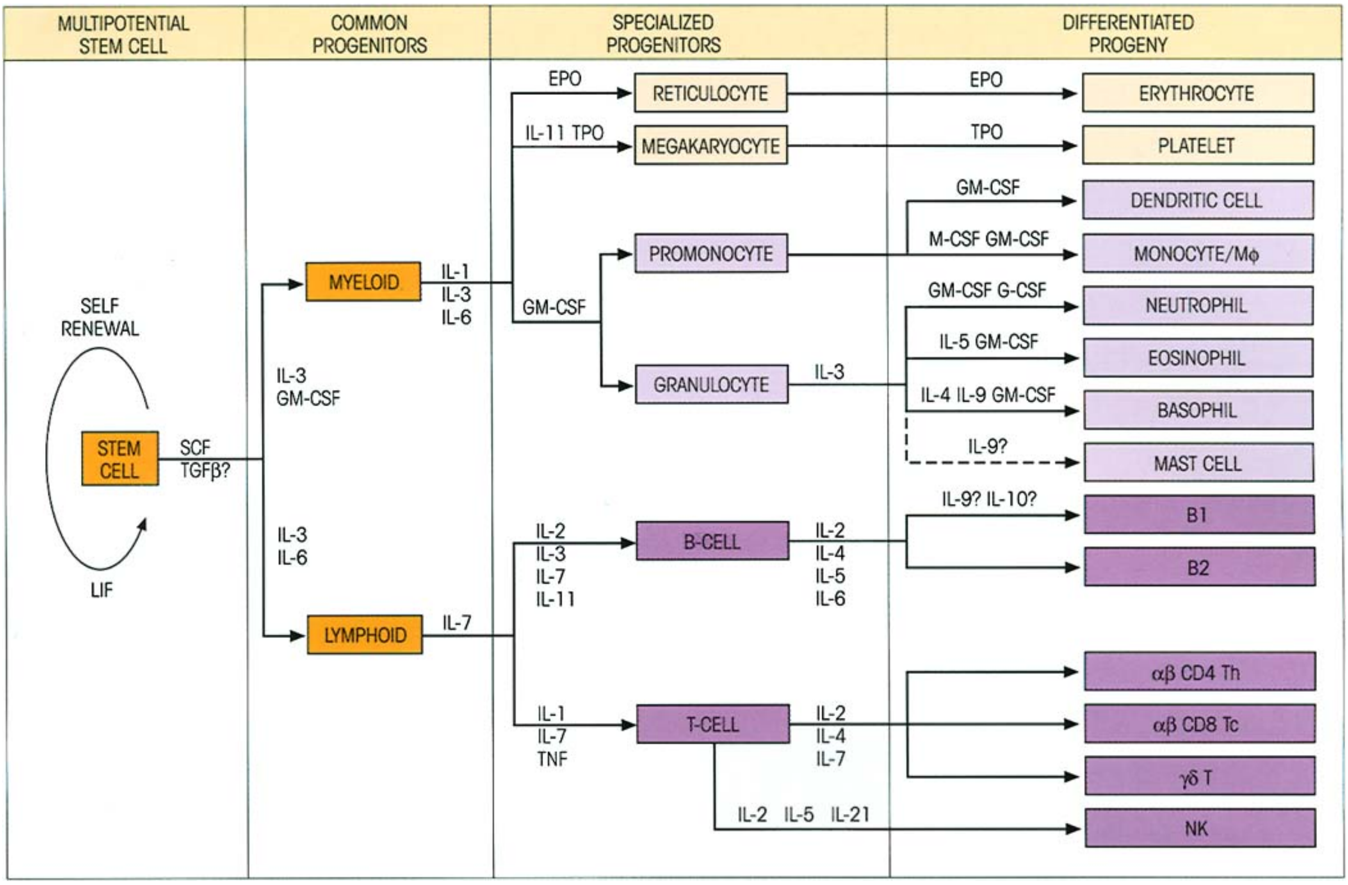

Figure 12.1. The multipotential hematopoietic stem cell and its progeny which differentiate under the influence of a series of soluble growth factors within the microenvironment of the bone marrow. The expression of various nuclear transcription factors directs the differentiation process. For example, the *Ikaros* gene encodes a zinc-fingered transcription factor critical for driving the development of a common myeloid/lymphoid precursor into a lymphoid-restricted progenitor giving rise to T-, B- and NK cells. SCF, stem cell factor; LIF, leukemia inhibitory factor; IL-3, interleukin-3, often termed the multi-CSF because it stimulates progenitors of platelets, erythrocytes, all the types of myeloid cells, and also the progenitors of B-, but not T-, cells; GM-CSF, granulocyte–macrophage colony-stimulating factor, so-called because it promotes the formation of mixed colonies of these two cell types from bone marrow progenitors either in tissue culture or on transfer to an irradiated recipient where they appear in the spleen; G-CSF, granulocyte colony-stimulating factor; M-CSF, monocyte colony-stimulating factor; EPO, erythropoietin; TPO, thrombopoietin; TNF, tumor necrosis factor; TGFβ, transforming growth factor β.

stem cells, and evidence was obtained to support an important role for TGFβ in the control of hematopoiesis. Similarly, the expression of Notch-1, our old friend NFκB and the wonderfully named Manic Fringe and Dishevelled-1 genes underscores the importance of the Notch signaling pathway in hematopoietic development. Finally, a large number of genes for cell adhesion molecules were present in the cDNA libraries, including several that implicate members of the semaphorin family in stem cell homing. In the human, CD34 is a marker of an extremely early cell but, again, there is some debate as to whether this identifies the holy pluripotent stem cell itself.

The stem cells differentiate within the microenvironment of sessile stromal cells which produce various growth factors such as IL-3, -4, -6 and -7, GM-CSF, and so on. The importance of this interaction between undifferentiated stem cells and the microenvironment which guides their differentiation is clearly shown by studies on mice homozygous for mutations at the *w* or the *sl* loci which, amongst other defects, have severe macrocytic anemia. Bone marrow stromal cells produce stem cell factor (SCF) which remains associated with the extracellular matrix and acts on primitive stem cells through a tyrosine kinase membrane receptor, c-kit (CD117). *sl*/*sl* mutants have normal stem cells but defective stromal production of SCF which can be corrected by transplantation of a normal spleen fragment; *w*/*w* mutant myeloid progenitors lack the c-kit surface receptor for SCF, and so can be restored by injection of normal bone marrow cells (figure 12.2).

Mice with *s*evere *c*ombined *i*mmuno*d*eficiency (SCID) provide a happy environment for fragments of

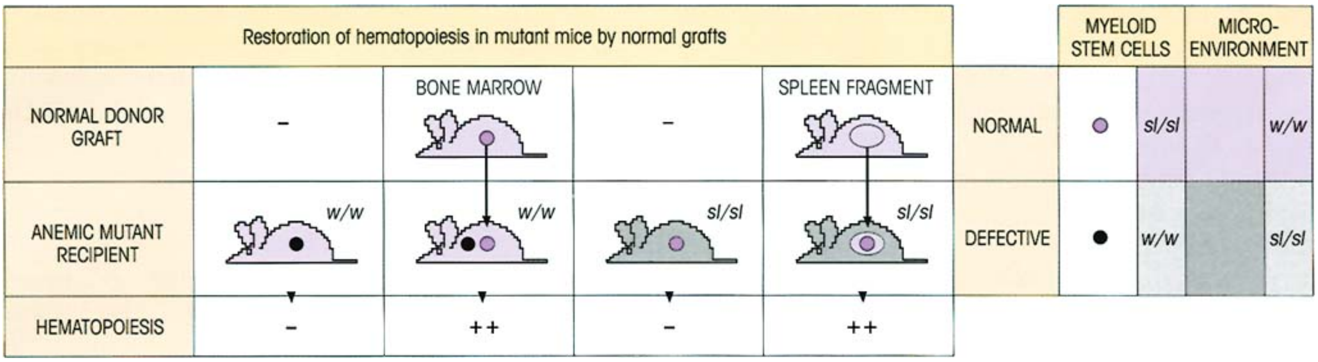

Figure 12.2. Hematopoiesis requires normal bone marrow stem cells differentiating in a normal microenvironment. The *w* locus codes for c-kit, a stem cell tyrosine kinase membrane receptor for the stem cell factor (SCF) encoded by the *sl* locus. Mice which are homozygous for mutant alleles at these loci develop severe macrocytic anemia which can be corrected by transplantation of appropriate normal cells. The experiments show that the *w/w* mutant lacks normal stem cells and the *sl/sl* mutant lacks the environmental factor needed for their development.

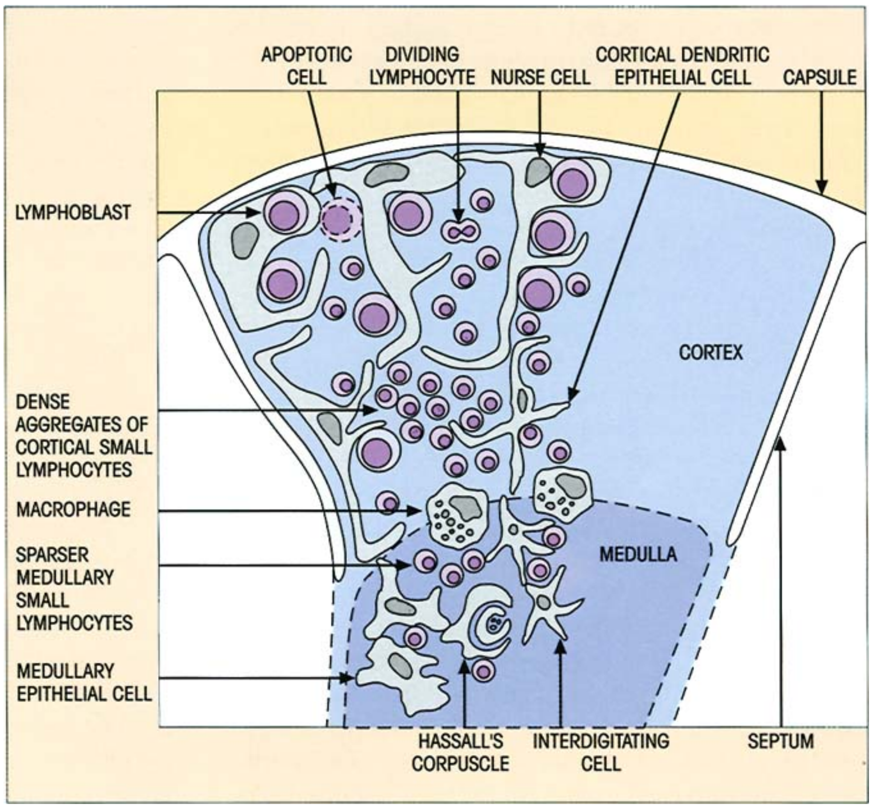

Figure 12.3. Cellular features of a thymus lobule. See text for description. (Adapted from Hood L.E., Weissman I.L., Wood W.B. & Wilson J.H. (1984) *Immunology*, 2nd edn, p. 261. Benjamin Cummings, California.)

human fetal liver and thymus which, if implanted contiguously, will produce formed elements of the blood for 6–12 months. Fetal liver provides a source of hematopoietic stem cells; but what is the role of the thymus?

THE THYMUS PROVIDES THE ENVIRONMENT FOR T-CELL DIFFERENTIATION

The thymus is organized into a series of lobules based upon meshworks of epithelial cells derived embryologically from an outpushing of the gut endoderm of the third pharyngeal pouch and which form well-defined cortical and medullary zones (figure 12.3). This framework of epithelial cells provides the microenvironment for T-cell differentiation. There are subtle interactions between the extracellular matrix proteins and a variety of integrins on different lymphocyte subpopulations produced by differential splicing and post-translational glycosylation; current musings are that the expression of these integrins, together with chemokine and chemokine receptor expression (cf. table 10.3), plays a role in the homing of progenitors to, and their subsequent migration within, the thymus. In addition, the epithelial cells produce a series of peptide hormones which mostly seem capable of promoting the appearance of T-cell differentiation markers and a variety of T-cell functions on culture with bone marrow cells *in vitro*. Several have been well characterized and sequenced, including thymulin, thymosin α_1, *t*hymic *h*umoral *f*actor (THF) and thymopoietin (and its active pentapeptide thymopontin, TP-5). Of these, only thymulin is of exclusively thymic origin. This zinc-dependent nonapeptide tends to normalize the balance of immune responses: it restores antibody avidity and antibody production in aged mice and yet stimulates suppressor activity in animals with autoimmune hemolytic anemia induced by cross-reactive rat red cells (cf. p. 416). Thymulin may be looked upon as a true hormone, secreted by the thymus in a regulated fashion and acting at a distance from the thymus as a fine physiological immunoregulator contributing to the maintenance of T-cell subset homeostasis.

Specialized large epithelial cells in the outer cortex, known as 'nurse' cells, are associated with large numbers of lymphocytes which lie within pockets produced by the long membrane extensions of these epithelial cells. The epithelial cells of the deep cortex have branched dendritic processes, rich in class II MHC, and connect through desmosome cell junctions to form a network through which cortical lymphocytes must pass on their way to the medulla (figure 12.3). The cortical lymphocytes are densely packed compared with those in the medulla, many are in division and large numbers of them are undergoing apoptosis. On their way to the medulla, the lymphocytes pass a cordon of 'sentinel' macrophages at the corticomedullary junction. A number of bone marrow-derived interdigitating dendritic cells are present in the medulla and the epithelial cells have broader processes than their cortical counterparts and express high levels of both class I and class II MHC. Whorled keratinized epithelial cells in the medulla form the highly characteristic Hassall's corpuscles beloved of histopathology examiners. These structures may serve as a disposal system for dying thymocytes and are the only location where apoptotic cells are found in the medulla. The presence of HLA-DO and HLA-DM, molecules which mediate intracellular peptide exchange, in the epithelial cells which ring the Hassall's corpuscles suggest that they may also play a role in activation and/or tolerance of mature thymocytes, particularly given that these epithelial cells also express the peptide-presenting HLA-DR molecule.

A fairly complex relationship with the nervous system awaits discovery; the thymus is richly innervated with both adrenergic and cholinergic fibers, while the neurotransmitters oxytocin, vasopressin and neurophysin are synthesized endogenously by subcapsular, perivascular and medullary epithelial cells and nurse cells. Acute stress leads to an indecently rapid loss of cortical thymocytes and an increase in epithelial cells expressing both cortical and medullary markers—surely intrathymic epithelial stem cells? The destruction of cortical thymocytes is at least partly due to the cytolytic action of steroids, the relative invulnerability of the medullary lymphocytes being attributable to their possession of a 20α-hydroxyl steroid dehydrogenase. The distinctive nature of the two main compartments in the gland is emphasized by the selective atrophy induced by a number of agents; thus the primary target of organotin is the immature cortical thymocyte. Dioxin interacts with a receptor on cortical epithelial cells, while the immunosuppressive drug cyclosporin A causes atrophy of all the medullary elements, thereby blocking differentiation of cortical to medullary thymocytes.

In the human, thymic involution commences within the first 12 months of life, reducing by around 3% a year to middle age and by 1% thereafter. The size of the organ gives no clue to these changes because there is replacement by adipose tissue. In a sense, the thymus is progressively disposable because, as we shall see, it establishes a long-lasting peripheral T-cell pool which enables the host to withstand loss of the gland without catastrophic failure of immunological function, witness the minimal effects of thymectomy in the adult compared with the **dramatic influence in the neonate** (Milestone 12.1). Nevertheless, the adult thymus retains a residue of corticomedullary tissue containing a normal range of thymocyte subsets with a broad spectrum of TCR gene rearrangements. Adult patients receiving either T-cell-depleted bone marrow or peripheral blood hematopoietic stem cells following ablative therapy are able to generate new naive T-cells at a rate that is inversely related to the age of the individual. These observations establish that new T-cells can be generated in adult life, either in the thymus or in the still mysterious 'extrathymic' sites that have been proposed as additional locations for T-cell differentiation.

Bone marrow stem cells become immunocompetent T-cells in the thymus

The evidence for this comes from experiments on the reconstitution of irradiated hosts. An irradiated animal is restored by bone marrow grafts through the

Milestone 12.1 — The Immunological Function of the Thymus

Ludwig Gross had found that a form of mouse leukemia could be induced in low-leukemia strains by inoculating filtered leukemic tissue from high-leukemia strains provided that this was done in the immediate neonatal period. Since the thymus was known to be involved in the leukemic process, Jacques Miller decided to test the hypothesis that the Gross virus could only multiply in the neonatal thymus by infecting neonatally thymectomized mice of low-leukemia strains. The results were consistent with this hypothesis but, strangely, animals of one strain died of a wasting disease which Miller deduced could have been due to susceptibility to infection, since fewer mice died when they were moved from the converted horse stables which served as an animal house to 'cleaner' quarters.

Autopsy showed the animals to have atrophied lymphoid tissue and low blood lymphocyte levels, and Miller therefore decided to test their immunocompetence before the onset of wasting disease. To his astonishment, skin grafts, even from rats (figure M12.1.1) as well as from other mouse strains, were fully accepted. These phenomena were not induced by thymectomy later in life and, in writing up his preliminary results in 1961 (Miller J.F.A.P., *Lancet* **ii**, 748), Miller opined that 'during embryogenesis the thymus would produce the originators of immunologically competent cells, many of which would have migrated to other sites at about the time of birth'. All in all a superb example of the scientific method and its application by a top-flight scientist.

Figure M12.1.1. Acceptance of a rat skin graft by a mouse which had been neonatally thymectomized.

immediate restitution of granulocyte precursors; in the longer term, also through reconstitution of the T- and B-cells destroyed by irradiation. However, if the animal is thymectomized before irradiation, bone marrow cells will not reconstitute the T-lymphocyte population (cf. figure 7.13).

By day 11–12 in the mouse embryo, lymphoblastoid stem cells from the bone marrow begin to colonize the periphery of the epithelial thymus rudiment. If the thymus is removed at this stage and incubated in organ culture, a whole variety of mature T-lymphocytes will be generated. This is not seen if 10-day thymuses are cultured and shows that the lymphoblastoid colonizers give rise to the immunocompetent small lymphocyte progeny.

T-CELL ONTOGENY

Differentiation is accompanied by changes in surface markers

The incoming thymic lymphoid progenitors express a number of chemokine receptors and are attracted to the thymus by one or more of the numerous chemokines secreted by the thymic stromal cells—which are the critical ones is yet to be established. These progenitors express CD34 and the enzyme *t*erminal *d*eoxynucleotidyl *t*ransferase (TdT) (figure 12.4), which is involved in the insertion of nucleotide sequences at the N-terminal region of *D* and *J* variable region segments to increase diversity of the T-cell receptors (TCRs) (cf. p. 65). They also express high levels of the adhesion molecule CD44 and the stem cell factor receptor (c-kit, CD117) (p. 222), but their lack of both CD4 and CD8 designates them as **double-negative** thymocytes. They express high levels of Notch molecules. These cell surface proteins provide signals at several key points during thymocyte differentiation. Indeed, they are thought to be necessary for commitment to the T-cell lineage, T-cell development being severely impaired in *Notch*$^{-/-}$ knockout mice. Under the influence of IL-1 and TNF, the progenitors differentiate into prothymocytes, committed to the T-lineage, and these now undergo IL-7-mediated proliferation to form a population of CD44^{-}CD117^{-} pre-T-cells. At this stage, the cells begin to express various TCR chains and are then expanded, ultimately synthesizing CD3, the invariant signal transducing complex of the TCR, and becoming **double positive** for CD4$^+$, CD8$^+$, the markers of the helper and cytotoxic subsets respectively. Finally, again under the guiding hand of chemokines, the cells traverse the corticomedullary junction to the medulla as the CD4 and CD8 markers segregate in parallel with differentiation into separate immunocompetent populations of **single-positive CD4$^+$ T-helpers** and **CD8$^+$ cytotoxic T-cell precursors** (figure 12.4). Notch-1 signaling is involved in the maturation of both of these subsets, with the Notch-1 ligands Jagged-1, Jagged-2 and δ-like-1 being expressed on thymic epithelial cells in a highly regulated way. The γδ cells remain double negative, i.e. CD4$^-$8$^-$, except for a small subset which express CD8.

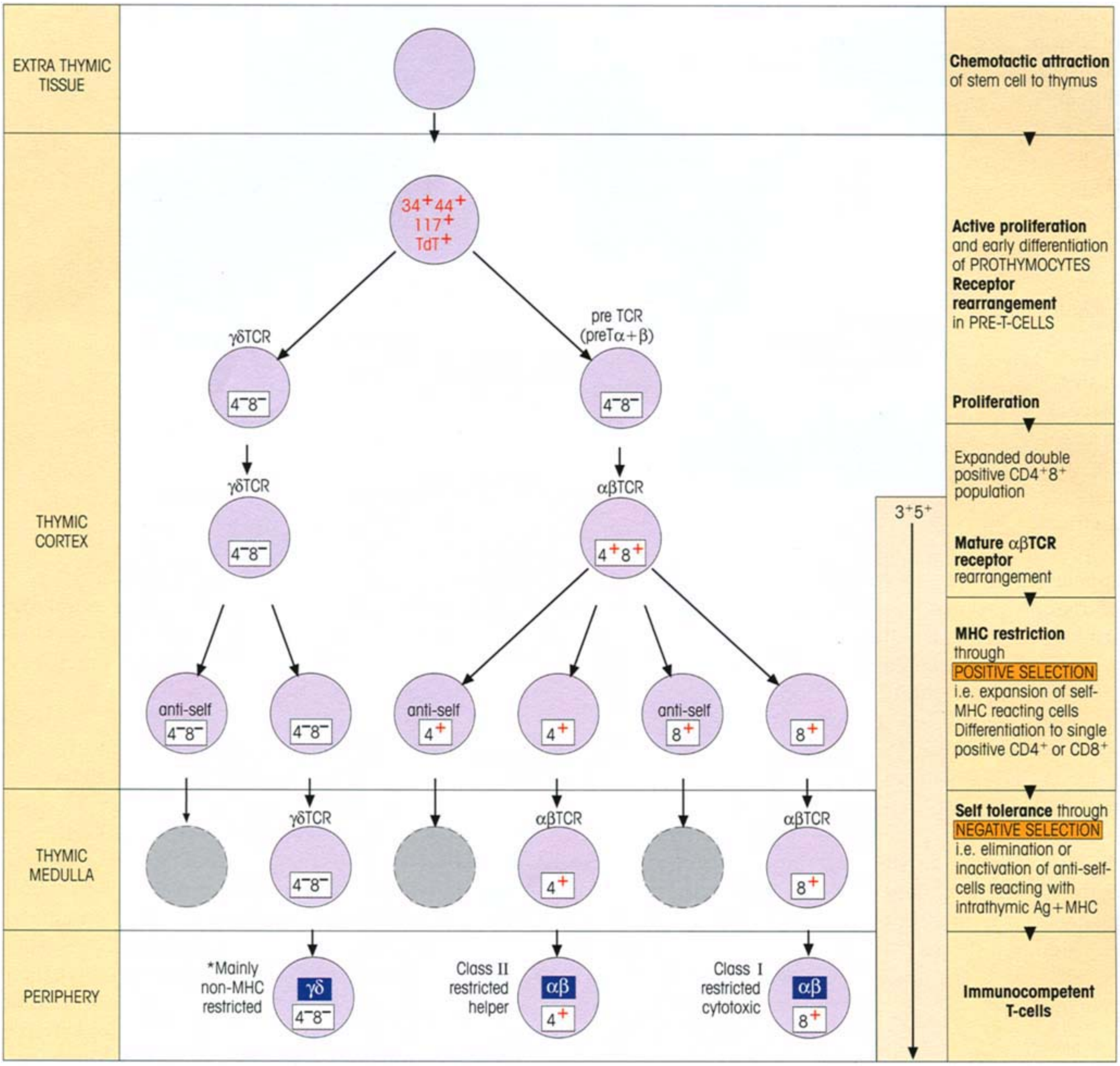

Figure 12.4. Differentiation of T-cells within the thymus. Numbers refer to CD designation. TdT, terminal deoxynucleotidyl transferase. Negatively selected cells in gray. The diagram is partly simplified for the sake of clarity. Autoreactive cells with specificity for self-antigens not expressed in the thymus may be tolerized by extrathymic peripheral contact with antigen (dashed circles). *γδ cells mainly appear to recognize antigen directly, in a manner analogous to the antibody molecule on B-cells, although some may be restricted by nonclassical MHC class Ib or by classical MHC class II. The relatively primitive NK-T cells bearing an αβ TCR and the NK1.1 marker (cf. p. 102) are often restricted by CD1, although some are restricted by classical MHC molecules.

The factors which determine whether the double-positive cells become CD4 or CD8 cells in the thymus are still not fully established. Two major scenarios have been put forward. The **stochastic** hypothesis suggests that expression of either CD4 or CD8 is randomly switched off, whereas the **instructive** hypothesis declares that interaction of the TCR with MHC–peptide results in signals which instruct the T-cells to become either $CD4^+$ class II-restricted cells or $CD8^+$ class I-restricted cells. The weight of the evidence currently available would seem to support the instructive model, whilst not entirely excluding the possibility of a contribution from stochastic events. Thus it appears that the strength of the $p56^{lck}$ signal (see p. 166) correlates with lineage choice and is determined by the duration of TCR engagement during the double-positive stage of thymocyte differentiation, a stronger cumulative signal favoring the generation of CD4 cells (figure 12.5).

Receptor rearrangement

The rearrangement of *V*-, *D*- and *J*-region genes required to generate the TCR (see p. 62) has not yet taken place at the prothymocyte stage. The gene for TdT and the recombinase activator genes, *RAG-1* and *RAG-2*, are transcribed at the pre-T stage and, by day 15, cells with the γδ TCR can be detected in the mouse thymus followed soon by the appearance of a '**pre-TCR**' version of the αβ TCR. Notch-1 signals (again!) appear to play a role in αβ versus γδ lineage commitment, although the details are still being worked out.

The development of αβ receptors

The *Vβ* is first rearranged in the double-negative $CD4^-8^-$ cells and associates with an invariant pre-α chain, pTα, to form a single 'pre-TCR' (figure 12.4). Although the pre-TCR mediates feedback inhibition on further TCR *Vβ* gene rearrangement, the extracellular domain of the pTα is not required for this inhibition, suggesting that there may not be an extracellular ligand for pTα, as had previously been assumed. A cysteine just inside the membrane on the cytoplasmic tail becomes palmitoylated and it may be this process that recruits pTα into lipid rafts containing the $p56^{lck}$ signaling molecule. The pre-T-cells undergo a frenetic burst of proliferation controlled by thymic epithelial cells and fibroblasts, producing double-positive $CD4^+8^+$ cells. Further development requires rearrangement of the *Vα* gene segments so allowing formation of the mature αβ TCR. The cells are now ready for subsequent bouts of positive and negative receptor editing as will be discussed shortly.

Rearrangement of the *Vβ* genes on the sister chromatid is suppressed following the expression of the pre-TCR (remember each cell contains two chromosomes for each α and β cluster). Thus each cell only expresses a single TCR β chain and the process by which the homologous genes on the sister chromatid are suppressed is called **allelic exclusion** (cf. p. 240). The α chains appear not to always be allelically excluded, so that some T-cells may have two antigen-specific receptors, each with their own α chain but sharing a common β chain. It is not clear, however, whether both receptors can be functional.

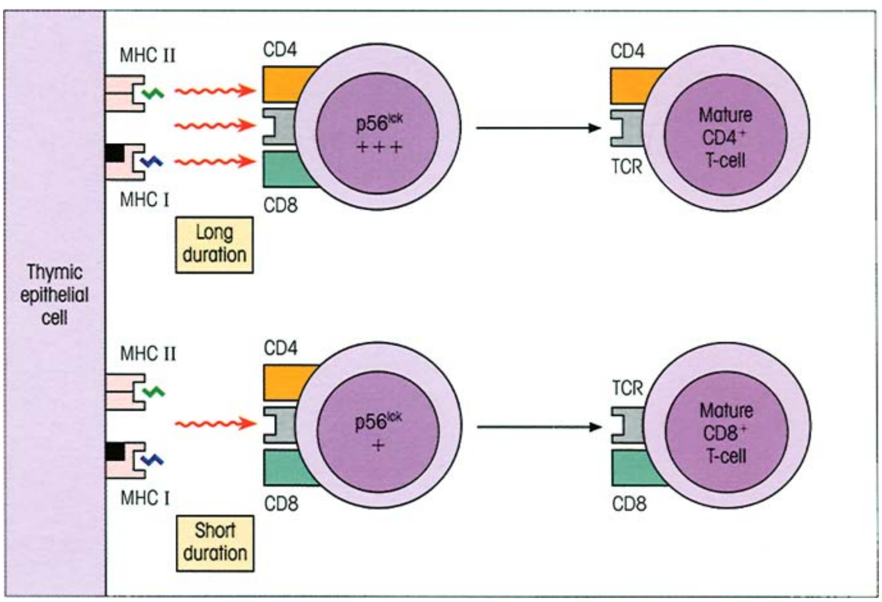

Figure 12.5. CD4 versus CD8 lineage commitment. In this model, the duration of signaling through the TCR and coreceptors on the double-positive ($CD4^+8^+$) thymocyte determines the intensity of the $p56^{lck}$ signal, which in turn instructs the thymocyte to become either a CD4 T-cell if there are sustained $p56^{lck}$ signals or a CD8 T-cell if the signals are less intense.

The development of γδ receptors

Unlike the αβ TCR, the γδ TCR in many cases seems to be able to bind directly to antigen without the necessity for antigen presentation by MHC or MHC-like molecules, i.e. it recognizes antigen directly in a manner similar to antibody. The γδ lineage does not produce a 'pre-receptor' and mice expressing rearranged *γ* and *δ* transgenes do not rearrange any further *γ* or *δ* gene segments, indicating allelic exclusion of sister chromatid genes.

γδ T-cells in the mouse, unlike the human, predominate in association with epithelial cells. A curious feature of the cells leaving the fetal thymus is the restriction in *V* gene utilization. Thus virtually all of the first wave of fetal γδ cells express the *Vγ5* gene and colonize the skin; the second wave use the same *δ* gene combination, but a different *γ V–J* pair utilizing *Vγ6*, and they seed the uterus in the female. In adult life, there is far more receptor diversity due to a high degree of junctional variation (cf. p. 65), although the intraepithelial cells in the intestine (*Vγ4*) and those in encapsulated lymphoid tissue (*Vγ4*, *Vγ1*, *Vγ2*) are again restricted with respect to *V* gene usage.

The Vγ set in the skin readily proliferates and secretes IL-2 on exposure to heat-shocked keratinocytes, implying a role in the surveillance of trauma signals. The Vγ4 cells in peripheral lymphoid tissue respond well to the tuberculosis antigen PPD ('purified protein derivative') and to conserved residues 180–196 from mycobacterial and self-heat-shock protein hsp65. However, evidence from γδ TCR knockout mice suggests that overall, in the adult, γδ T-cells may make a minor contribution to pathogen-specific protection. It has therefore been proposed that their primary role may be in the regulation of αβ T-cells, with most γδ T-cells biased towards a Th1 cytokine secretion pattern.

Two major γδ subsets predominate in the human, Vγ9,Vδ2 and Vγ1,Vδ2. The Vγ9 set rises from 25% of the total γδ cells in cord blood to around 70% in adult blood; at the same time, the proportion of Vγ1 falls from 50% to less than 30%. The majority of the Vγ9 set have the activated memory phenotype CD45RO, probably as a result of stimulation by common ligands for the Vγ9,Vδ2 TCR, such as components of mycobacteria, *Plasmodium falciparum* and the superantigen staphylococcal enterotoxin A. Vγ9 subsets of extremely limited junctional diversity were observed in the blood and bronchoalveolar lavage of two patients with sarcoidosis, a granulomatous disease with mycobacterial involvement. The conclusion that these two major γδ subpopulations are selected by powerful antigens seems inescapable.

Cells are positively selected for self-MHC restriction in the thymus

The ability of T-cells to recognize antigenic peptides in association with self-MHC is developed in the thymus. If an (H-2^k×H-2^b) F1 animal is sensitized to an antigen, the primed T-cells can recognize that antigen on presenting cells of either *H*-2^k or *H*-2^b haplotype, i.e. they can use either parental haplotype as a recognition restriction element. However, if bone marrow cells from the (H-2^k×H-2^b) F1 are used to reconstitute an irradiated F1 which had earlier been thymectomized and given an H-2^k thymus, the subsequently primed T-cells can only recognize antigens in the context of H-2^k, not of H-2^b (figure 12.6). Thus it is **the phenotype of the**

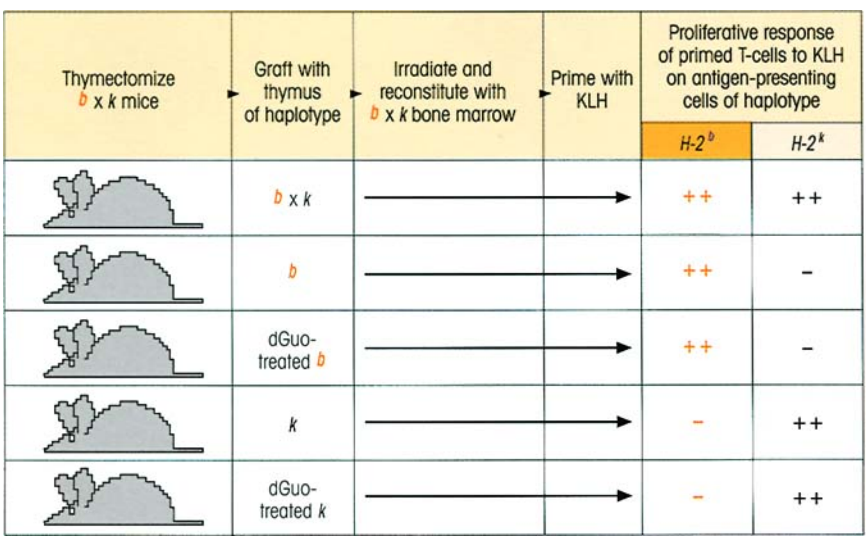

Figure 12.6. Imprinting of H-2 T-helper restriction by the haplotype of the thymus. Host mice were F1 crosses between strains of haplotype *H*-2^b and *H*-2^k. They were thymectomized and grafted with 14-day fetal thymuses, irradiated and reconstituted with F1 bone marrow. After priming with the antigen keyhole limpet hemocyanin (KLH), the proliferative response of lymph node T-cells to KLH on antigen-presenting cells of each parental haplotype was assessed. In some experiments, the thymus lobes were cultured in deoxyguanosine (dGuo), which destroys intrathymic cells of macrophage/dendritic cell lineage, but this had no effect on positive selection. (From Lo D. & Sprent J. (1986) *Nature* **319**, 672.)

thymus that imprints H-2 restriction on the differentiating T-cells.

It will also be seen in figure 12.6 that incubation of the thymus graft with deoxyguanosine, which destroys the cells of macrophage and dendritic cell lineage, has no effect on imprinting, suggesting that this function is carried out by epithelial cells. Confirmation of this comes from a study showing that lethally irradiated H-2^k mice, reconstituted with (b×k) F1 bone marrow and then injected intrathymically with an H-2^b thymic epithelial cell line, developed T-cells restricted by the *b* haplotype. The epithelial cells are rich in surface MHC molecules and the current view is that double-positive ($CD4^+8^+$) T-cells bearing receptors which recognize self-MHC on the epithelial cells are positively selected for differentiation to $CD4^+8^-$ or $CD4^-8^+$ single-positive cells. The evidence for this comes largely from studies in transgenic mice. Since this is a very active area, we would like to cite some experimental examples; nonprofessionals may need to hang on to their haplotypes, put on their ice-packs and concentrate.

One highly sophisticated study starts with a cytotoxic T-cell clone raised in H-2^b females against male cells of the same strain. The clone recognizes the male antigen, H-Y, and this is seen in association with the H-$2D^b$ self-MHC molecules, i.e. it reacts with the H-2^b/Y complex. The α and β chains for the T-cell receptor of this clone are now introduced as transgenes into SCID mice which lack the ability to rearrange their own germ-line variable region receptor genes; thus the only TCR which could possibly be expressed is that encoded by the transgenes, provided of course that we are looking at females rather than males, in whom the clone would be eliminated by self-reactivity. If the transgenic SCID females bear the original *H-2^b* haplotype (e.g. F1 hybrids between *b*×*d* haplotypes), then the anti-H-2^b/Y receptor is amply expressed on $CD8^+$ cytotoxic precursor cells (table 12.1a), whereas H-2^d transgenics lacking H-2^b produce only double $CD4^+8^+$ thymocytes with no single $CD4^+8^-$ or $CD4^-8^+$ cells. Thus, as $CD4^+8^+$ cells express their TCR transgene, they only differentiate into $CD8^+$ immunocompetent cells if they come into contact with thymic epithelial cells of the MHC haplotype recognized by their receptor. We say that such self-recognizing thymocytes are being **positively selected**. Positive intracellular events accompany the positive selection process since the protein tyrosine kinases fyn and lck are activated in double-positive $CD4^+8^+$ thymocytes maturing to single-positive $CD8^+$ cells in the *b* haplotype background, but are low in cells which fail to differentiate into mature cells in the nonselective *d* haplotype.

Table 12.1. Positive and negative selection in SCID transgenic mice bearing the αβ receptors of an H-$2D^b$ T-cell clone cytotoxic for the male antigen H-Y, i.e. the clone is of *H-2^b* haplotype and is female anti-male. (a) The only T-cells are those bearing the already rearranged transgenic TCR, since SCID mice cannot rearrange their own *V* genes. The clones are only expanded beyond the $CD4^+8^+$ stage when positively selected by contact with the MHC haplotype (*H-2^b*) recognized by the original clone from which the transgene was derived. Also, since the TCR recognized class I, only $CD8^+$ cells were selected. (b) When the anti-male transgenic clone is expressed on intrathymic T-cells in a male environment, the strong engagement of the TCR with male antigen-bearing cells eliminates them. (Based on data from von Boehmer H. *et al.* (1989) In Melchers F. *et al.* (eds) *Progress in Immunology* 7, p. 297. Springer-Verlag, Berlin.)

Phenotype of thymocytes	a Positive selection: Haplotype of transgenic females		b Negative selection: Transgenic H-2^b mice	
	H-$2^{b/d}$	*H-$2^{d/d}$*	Males	Females
$CD4^-8^-$ TCR^-	+	++	+++	+
$CD4^+8^+$ $TCR^{\pm}$	++	+	–	+++
$CD4^-8^+$ TCR^{++}	+	–	–	+
$CD4^+8^-$ TCR^{++}	–	–	–	–

+, crude measure of the relative numbers of T-cells in the thymus having the phenotype indicated.

In another example, genes encoding an αβ receptor from a T-helper clone (2B4), which responds to moth cytochrome *c* in association with the class II molecule H-2Eα^k,β^b (remember H-2E has an α and β chain), are transfected into H-2^k and H-2^b mice. For irrelevant reasons, H-2^k mice express the H-2E molecule on the surface of their antigen-presenting cells, but H-2^b do not. In the event, the frequency of circulating $CD4^+$ T-cells bearing the 2B4 receptor was 10 times greater in the H-2^k relative to H-2^b strains, again speaking for positive selection of double-positive thymocytes which recognize their own thymic MHC. In a further twist to the story, positive selection only occurred in mice manipulated to express H-2E on their cortical rather than their medullary epithelial cells, showing that this differentiation step is effected before the developing thymocytes reach the medulla. ('Read it again Sam' as Humphrey Bogart might have said!)

T-CELL TOLERANCE

The induction of immunological tolerance is necessary to avoid self-reactivity

In essence, lymphocytes recognize foreign antigens through complementarity of shape mediated by the intermolecular forces we have described previously (see p. 85). To a large extent the building blocks used to form microbial and host molecules are the same, and so it is the assembled shapes of *self* and *nonself* mole-

Milestone 12.2—The Discovery of Immunological Tolerance

Over 40 years ago, Owen made the intriguing observation that nonidentical (dizygotic) twin cattle, which shared the same placental circulation and whose circulations were thereby linked, grew up with appreciable numbers of red cells from the other twin in their blood; if they had not shared the same circulation at birth, red cells from the twin injected in adult life would have been rapidly eliminated by an immunological response. From this finding, Burnet and Fenner conceived the notion that potential antigens which reach the lymphoid cells during their developing immunologically immature phase can in some way specifically suppress any future response to that antigen when the animal reaches immunological maturity. This, they considered, would provide a means whereby unresponsiveness to the body's own constituents could be established and thereby enable the lymphoid cells to make the important distinction between 'self' and 'nonself'. On this basis, any foreign cells introduced into the body during immunological development should trick the animal into treating them as 'self'-components in later life, and the studies of Medawar and his colleagues have shown that **immunological tolerance**, or unresponsiveness, can be artificially induced in this way. Thus neonatal injection of CBA mouse cells into newborn A strain animals suppresses their ability to reject a CBA graft immunologically in adult life (figure M12.2.1). Tolerance can also be induced with soluble antigens; for example, rabbits injected with bovine serum albumin without adjuvant at birth fail to make antibodies on later challenge with this protein.

Persistence of antigen is required to maintain tolerance. In Medawar's experiments, the tolerant state was long lived because the injected CBA cells survived and the animals continued to be chimeric (i.e. they possessed both A and CBA cells). With nonliving antigens, such as soluble bovine serum albumin, tolerance is gradually lost; the most likely explanation is that, in the absence of antigen, newly recruited immunocompetent cells which are being generated throughout life are not being rendered tolerant. Since recruitment of newly competent T-lymphocytes is drastically curtailed by removal of the thymus, it is of interest to note that the tolerant state persists for much longer in thymectomized animals.

The vital importance of the experiments by Medawar and his team was their demonstration that a state of immunological tolerance can result from exposure to an antigen. Recent studies, however, suggest that the concept of a *neonatal window* for tolerance induction is more apparent than real and stems from the relatively low number of peripheralized immunocompetent T-cells, which do not differ in behavior from resting T-cells in the adult in their tolerizability or capacity for an immune response (see papers in *Science* (1996) **271**, 1723, 1726 and 1728), albeit that resting T-cells are more readily tolerizable than memory cells. As will be discussed in the text, there is a window of susceptibility to clonal deletion of self-reacting T-lymphocytes at an immature phase in their ontogenic development within the thymus (and in the case of B-cells within the bone marrow).

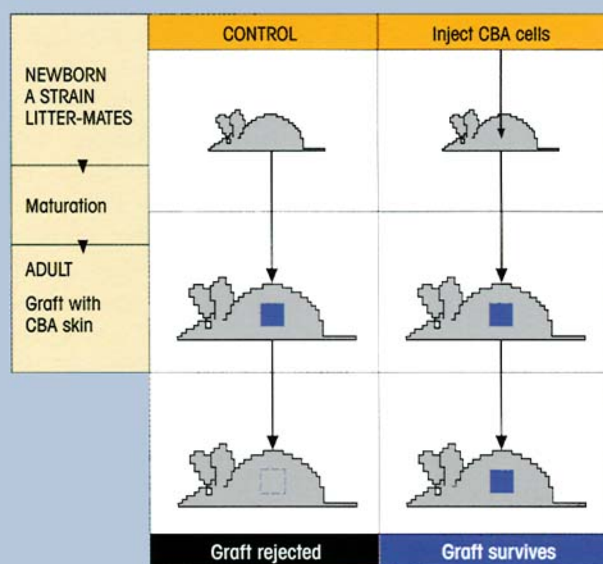

Figure M12.2.1. Induction of tolerance to foreign CBA skin graft in A strain mice by neonatal injection of antigen. The effect is antigen specific since the tolerant mice can reject third-party grafts normally. (After Billingham R., Brent L. & Medawar P.B. (1953) *Nature* **172**, 603.)

cules which must be discriminated by the immune system if potentially disastrous autoreactivity is to be avoided. The restriction of each lymphocyte to a single specificity makes the job of establishing self-tolerance that much easier, simply because it just requires a mechanism which functionally deletes self-reacting cells and leaves the remainder of the repertoire unscathed. The most radical difference between self and nonself molecules lies in the fact that, in early life, the developing lymphocytes are surrounded by self and normally only meet nonself antigens at a later stage and then within the context of the adjuventicity and cytokine release usually associated with infection. With its customary efficiency, the blind force of evolution has exploited these differences to establish the mechanisms of **immunological tolerance to host constituents** (Milestone 12.2).

Self-tolerance can be induced in the thymus

Since developing T-cells are to be found in the thymus, one might expect this to be the milieu in which exposure to self-antigens on the surrounding cells would induce tolerance. The expectation is reasonable. If stem cells in bone marrow of *H-2^k^* haplotype are cultured with fetal thymus of H-2^d origin, the maturing cells become tolerant to H-2^d, as shown by their inability to give a mixed lymphocyte proliferative response when cultured with stimulators of H-2^d phenotype; third-party responsiveness is not affected. Further experiments with deoxyguanosine-treated thymuses showed that the cells responsible for tolerance induction were deoxyguanosine-sensitive, bone marrow-derived macrophages or dendritic cells which are abundant at the corticomedullary junction (table 12.2).

Intrathymic clonal deletion leads to self-tolerance

There seems little doubt that self-reactive T-cells can be physically deleted within the thymus. If we look at the experiment in table 12.1b, we can see that SCID males bearing the rearranged transgenes coding for the αβ receptor reacting with the male H-Y antigen do not possess any immunocompetent thymic cells expressing this receptor, whereas the females which lack H-Y do. Thus, when the developing T-cells react with self-antigen in the thymus, they are deleted. In other words, self-reactive cells undergo a **negative selection** process in the thymus. A similar phenomenon is seen when the thymic cells bear certain self-components which act as superantigens (cf. p. 103) by reacting with a whole family of Vβ receptors through recognition of nonvariable structures on a Vβ segment. An example is the H-2E molecule which reacts with receptors belonging to the Vβ17a family; strains which cannot express H-2E because of a defect in the *Ea* gene possess mature T-cells utilizing Vβ17a, whereas strains which express H-2E normally delete their Vβ17a-positive T-cells. Likewise, mice of the *Mls^a^* genotype delete Vβ6-bearing cells, the *Mls* being a locus encoding a B-cell superantigen which induces strong proliferation in Vβ6 T-cells from a strain bearing a different *Mls* allele (cf. p. 104). Even exogenous superantigens, such as staphylococcal enterotoxin B which activates the Vβ3 and Vβ8 T-cell families in the adult, will eliminate these cells when incubated with early immature thymocytes. Even more enlightening is the fact that, under these circumstances, the Vβ3 and Vβ8 thymocytes can actually be seen to undergo apoptosis (cf. p. 19).

Table 12.2. Induction of tolerance in bone marrow stem cells by incubation with deoxyguanosine (dGuo)-sensitive macrophages or dendritic cells in the thymus. Clearly, the bone marrow cells induce tolerance to their own haplotype. Thus the thymic tolerance-inducing cells can be replaced by progenitors in the bone marrow inoculum (Jenkinson E.J., Jhittay P., Kingston R. & Owen J.J. (1985) *Transplantation* **39**, 331) or by adult dendritic cells from spleen, showing that it is the stage of differentiation of the immature T-cell rather than any special nature of the thymic antigen-presenting cell which leads to tolerance (Matzinger P. & Guerder S. (1989) *Nature* **338**, 74).

Bone marrow cells	Incubate with H-2^d thymus	Tolerance induction to *H-2* haplotype		
		k	*d*	*b*
k	Untreated	+	+	–
k	dGuo-treated	+	–	–
k + d	dGuo-treated	+	+	–

Factors affecting positive or negative selection in the thymus

It is established that engagement of TCR by the MHC–peptide complex on some type of antigen-presenting cell underlies both positive and negative selection. But how can the same MHC–peptide signal have two totally different outcomes? Well, positive and negative selection may occur at low and high degrees of TCR ligation, respectively. For example, high concentrations of antibody to the TCR induce apoptosis in thymocytes (figure 12.7), whereas low concentrations of anti-TCR do not. Furthermore, many examples have been published showing that the same peptide will induce positive selection at low concentration and negative selection at high concentration (see legend to figure 12.8). This has led to the avidity model, which postulates that a functionally low avidity interaction between T-cell and peptide–MHC involving a relatively low number of TCRs will positively select double-positive $CD4^+8^+$ thymocytes, while a high avidity interaction will lead to clonal deletion (figure 12.8). Since the overall avidity of the T-cell interaction will be *inter alia* a function of ligand density × TCR density × affinity, an increase in peptide concentration will increase ligand density and hence avidity. One problem will be immediately apparent to the discerning reader in that a given peptide ligand, giving a low avidity initial stimulus for positive selection, should give a negative signal as the thymocyte differentiates and the density of TCRs increases with the change from double- to single-positive cells. This has led to the suggestion that thymic cortical cells progressively desensitize the maturing thymocyte so that it resists the more powerful stimulus of the macrophages and medullary dendritic cells, which would otherwise induce apoptosis. Evidence is also accumulating that thymic epithelium can synthesize glucocorticoids, hormones classically associated with the adrenal gland. Maybe the *g*luco-corticoid-*i*nduced *l*eucine *z*ipper (GILZ) (cf. p. 215) protects the cells from a signal that would otherwise result in activation-induced cell death.

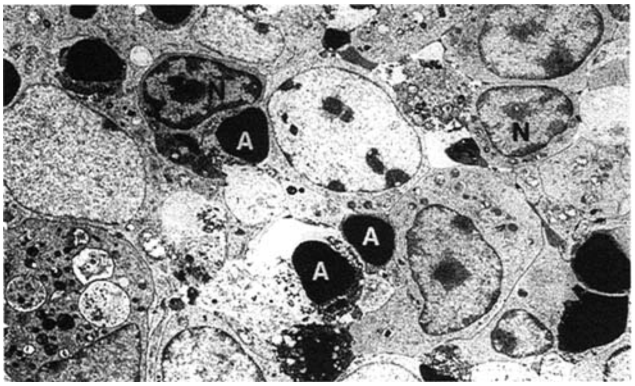

Figure 12.7. Electron micrograph of cells induced to undergo apoptosis in intact fetal thymus lobes after short-term exposure to anti-CD3. A and N indicate representative apoptotic and normal lymphocytes, respectively. Note the highly condensed state of the nuclei of the apoptotic lymphocytes. (Photograph kindly donated by Professor J.J.T. Owen, from Smith *et al.* (1989) *Nature* **337**, 181. Reproduced by permission from Macmillan Journals Ltd, London.)

To pause for a moment, we seem to be saying that engagement of the TCR of differentiating double-positive CD4^{+}8^{+} thymocytes with self-MHC on cortical epithelial cells leads to expansion and positive selection for clones which recognize self-MHC, perhaps with a whole range of affinities, but that engagement of the TCR with high affinity for self-MHC (+self-peptide) on bone marrow-derived medullary cells will lead to elimination and hence negative selection. Although still not fully worked out, there are also obvious differences in the biochemical pathways used for positive and negative signaling. Positive selection is cyclosporin A-sensitive and dependent on the Ras–MEK–ERK pathway (cf. p. 167), whereas negative selection is cyclosporin A-resistant and independent of this pathway. Different intensities of signaling from the TCR, and/or the types of coreceptor used, may influence which pathway is utilized. Let us finish on a cautionary note: the avidity model may be substantially correct but it could be an oversimplification. For instance, certain superantigens, which can cause clonal deletion of certain Vβ families, fail to expand them even at very low concentrations when the model would have indicated positive selection. This has spawned other models involving conformational changes, and given the complex interactions of peptides behaving as agonists, partial agonists and antagonists (cf. p. 170); the last word has not yet been spoken (not that it ever is in science!).

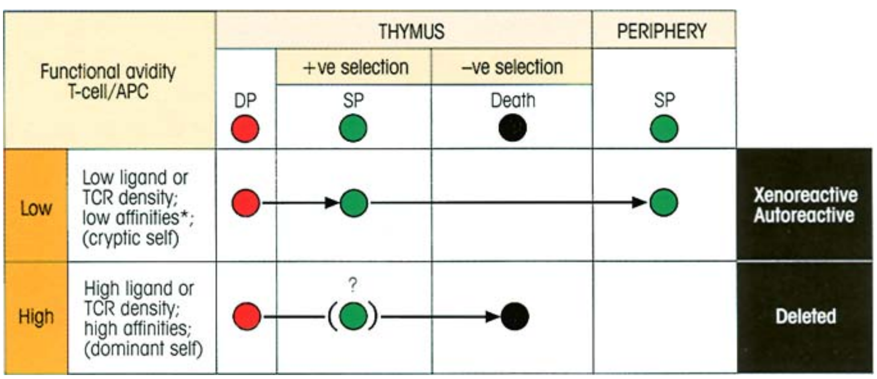

Figure 12.8. The avidity model of thymic positive and negative selection. It is postulated that a low avidity interaction between the T-cell and antigen-presenting cell (APC) will give positive selection and that high avidity will give deletion. DP, double-positive CD4^{+}8^{+}; SP, single-positive CD4^{+} or CD8^{+}; *refers to affinity of peptide for the MHC or of the MHC–peptide complex for the TCR.

When Tap-1 mutant mice (cf. p. 93) are mated with mice bearing the transgenes for the TCR specific for a complex of H-2D^{b} with an LCM virus peptide, the positive selection of the transgenic T-cells is impaired because of lack of MHC–peptide. However, low concentrations of the peptide added to fetal organ cultures of these mice selected the transgenic T-cells positively, while higher concentrations gave negative selection (Ljunggren H.G. & van Kaer L. (1995) *The Immunologist* **3**, 136). 'Cryptic self'-peptides (cf. p. 200) are presented at very low concentrations and will not delete potentially autoreactive clones, which may therefore escape to the periphery.